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Inventor(s)	Minami; Junya

Air conditioning indoor unit and air conditioner

Abstract

An air conditioning indoor unit that blows air, which has exchanged heat with a refrigerant flowing through a heat exchanger, into an air conditioning target space, includes: a liquid refrigerant pipe and a gas refrigerant pipe that are connected to the heat exchanger; a casing that accommodates the heat exchanger and that has an opening communicating with the air conditioning target space; a first shutoff valve in the liquid refrigerant pipe; a second shutoff valve in the gas refrigerant pipe; and a partition wall that separates a first space from a second space in the casing. The first shutoff valve and the second shutoff valve are in the first space. The second space communicates with the air conditioning target space via the opening.

Inventors:	**Minami; Junya (Osaka, JP)**
Applicant:	**DAIKIN INDUSTRIES, LTD.** (Osaka, JP)
Family ID:	**75848416**
Assignee:	**DAIKIN INDUSTRIES, LTD. (Osaka, JP)**
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2078295	12/1936	Thatcher	62/441	F24F 1/0083
2011/0192184	12/2010	Yamashita et al.	N/A	N/A
2012/0272672	12/2011	Morimoto et al.	N/A	N/A
2014/0238069	12/2013	Hayashi	N/A	N/A
2021/0231343	12/2020	Matsumoto et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102112815	12/2010	CN	N/A
105074359	12/2014	CN	F24F 3/065
209131012	12/2018	CN	N/A
1512919	12/2004	EP	N/A
2051017	12/2008	EP	F24F 1/0007
3569955	12/2018	EP	N/A
2566201	12/2018	GB	N/A
S57-077830	12/1981	JP	N/A
H03-061794	12/1990	JP	N/A
H08-178397	12/1995	JP	N/A
H1078273	12/1997	JP	N/A
H11-230565	12/1998	JP	N/A
2001-74264	12/2000	JP	N/A
2003-90557	12/2002	JP	N/A

2005-241050	12/2004	JP	N/A
2010-7998	12/2009	JP	N/A
2013-19621	12/2012	JP	N/A
2014-163548	12/2013	JP	N/A
2015-230136	12/2014	JP	N/A
2016-84946	12/2015	JP	N/A
2016084946	12/2015	JP	N/A
2019-45103	12/2018	JP	N/A
2004/020913	12/2003	WO	N/A
2011/099063	12/2010	WO	N/A
WO-2018011994	12/2017	WO	F24F 11/36
2018/131085	12/2017	WO	N/A
2019049746	12/2018	WO	N/A
2019111783	12/2018	WO	N/A

OTHER PUBLICATIONS

International Preliminary Report on Patentability issued in corresponding International Application No. PCT/JP2020/041102 mailed May 19, 2022 (6 pages). cited by applicant
International Search Report issued in corresponding International Application No. PCT/JP2020/041102 mailed Jan. 12, 2021 (5 pages). cited by applicant
Extended European Search Report issued in European Application No. 20885072.7, dated Nov. 17, 2022 (10 pages). cited by applicant

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Primary Examiner: Vazquez; Ana M

Attorney, Agent or Firm: Osha Bergman Watanabe & Burton LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation application of International Patent Application No. PCT/JP2020/041102, filed on Nov. 2, 2020, and claims priority to Japanese Patent Application No. 2019-201070, filed on Nov. 5, 2019, Japanese Patent Application No. 2019-217389, filed on Nov. 29, 2019, and Japanese Patent Application No. 2019-217390, filed on Nov. 29, 2019. The contents of these priority applications are incorporated herein by reference.

TECHNICAL FIELD
(1) The present disclosure relates to an air conditioning indoor unit and an air conditioner.
BACKGROUND
(2) As disclosed in Patent Literature 1 (JP 2018/011994 W), there is known an air conditioner in which a shutoff valve is provided in a refrigerant connection pipe connecting an air conditioning outdoor unit and an air conditioning indoor unit for preventing leakage of refrigerant.
SUMMARY
(3) An air conditioning indoor unit according to one or more embodiments blows air, which has exchanged heat with a refrigerant flowing through a heat exchanger, into an air conditioning target space. The air conditioning indoor unit includes a liquid refrigerant pipe and a gas refrigerant pipe connected to the heat exchanger, a casing, a first shutoff valve and a second shutoff valve, and a partition wall. The casing accommodates the heat exchanger. An opening communicating with the air conditioning target space is formed in the casing. The first shutoff valve and the second shutoff

valve are disposed in a first space in the casing. The first shutoff valve is disposed in the liquid refrigerant pipe. The second shutoff valve is disposed in the gas refrigerant pipe. The partition wall separates the first space and a second space. The second space is a space in the casing and communicates with the air conditioning target space via the opening.
(4) An air conditioner according to one or more embodiments includes an air conditioning indoor unit, an air conditioning heat source unit, and a shutoff valve device. The air conditioning indoor unit is installed in an air conditioning target space. The air conditioning heat source unit is connected to the air conditioning indoor unit via a liquid refrigerant pipe and a gas refrigerant pipe. The shutoff valve device includes a shutoff valve disposed in an attic space above a ceiling of an air conditioning target space. The shutoff valve includes at least one of a first shutoff valve disposed in the liquid refrigerant pipe and a second shutoff valve disposed in the gas refrigerant pipe.
(5) An air conditioner according to one or more embodiments includes an air conditioning indoor unit, an air conditioning heat source unit, and a shutoff valve device. The air conditioning indoor unit is installed in an air conditioning target space. The air conditioning indoor unit is a floor type. The air conditioning heat source unit is connected to the air conditioning indoor unit via a liquid refrigerant pipe and a gas refrigerant pipe. The shutoff valve device includes a shutoff valve disposed in an underfloor space below a floor of an air conditioning target space. The shutoff valve includes at least one of a first shutoff valve disposed in the liquid refrigerant pipe and a second shutoff valve disposed in the gas refrigerant pipe.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic configuration diagram of an air conditioner according to one or more embodiments.
(2) FIG. **2** is an external perspective view of an air conditioning indoor unit of the air conditioner of FIG. **1**.
(3) FIG. **3** is a schematic cross-sectional view of the air conditioning indoor unit attached to the ceiling, taken along line III-III in FIG. **2**.
(4) FIG. **4** is a bottom view schematically illustrating a schematic configuration of the air conditioning indoor unit in FIG. **2**, illustrating the air conditioning indoor unit from which a decorative panel, a bottom plate, and the like are removed.
(5) FIG. **5** is a schematic perspective view around a utilization heat exchanger for explaining a first space in which a first shutoff valve and a second shutoff valve are disposed.
(6) FIG. **6** is a schematic configuration diagram of an air conditioner including an air conditioning indoor unit according to Modification 1A.
(7) FIG. **7** is a schematic plan view for explaining an arrangement of devices inside an air conditioning indoor unit according to Modification 1C, in which a top panel of a casing is not shown.
(8) FIG. **8** is a schematic configuration diagram of an air conditioner according to one or more embodiments.
(9) FIG. **9** is a control block diagram of the air conditioner of FIG. **8**.
(10) FIG. **10**A is a diagram schematically illustrating installation states of a utilization unit and a shutoff valve device in a case where the utilization unit of the air conditioner in FIG. **8** is a wall-mounted type.
(11) FIG. **10**B is a diagram schematically illustrating installation states of the utilization unit and the shutoff valve device in a case where the utilization unit of the air conditioner in FIG. **8** is a floor type.

(12) FIG. **10**C is a diagram schematically illustrating installation states of the utilization unit and the shutoff valve device in a case where the utilization unit of the air conditioner in FIG. **8** is a ceiling suspended type.
(13) FIG. **11**A is a side view schematically illustrating a main body casing and an electric component box of the shutoff valve device.
(14) FIG. **11**B is a side view schematically illustrating a main body casing and an electric component box of another example of the shutoff valve device.
(15) FIG. **12** is a schematic configuration diagram of an air conditioner including a shutoff valve device according to Modification 2B.
(16) FIG. **13** is a schematic configuration diagram of an air conditioner including a shutoff valve device according to Modification 2C.
(17) FIG. **14** is a schematic configuration diagram of an air conditioner according to one or more embodiments.
(18) FIG. **15** is a control block diagram of the air conditioner of FIG. **14**.
(19) FIG. **16** is a diagram schematically illustrating installation states of a utilization unit and a shutoff valve device of the air conditioner in FIG. **14**.
(20) FIG. **17**A is a side view schematically illustrating a main body casing and an electric component box of the shutoff valve device.
(21) FIG. **17**B is a side view schematically illustrating a main body casing and an electric component box of another example of the shutoff valve device.
(22) FIG. **18** is a schematic configuration diagram of an air conditioner including a shutoff valve device according to Modification 3B.
(23) FIG. **19** is a schematic configuration diagram of an air conditioner including a shutoff valve device according to Modification 3C.

DETAILED DESCRIPTION

(24) Hereinafter, an air conditioning indoor unit and an air conditioner including the air conditioning indoor unit will be described with reference to the drawings.
(25) In the following description, for convenience of explanation, expressions such as up, down, left, right, front, and back may be used to describe directions and positional relationships. The directions indicated by these expressions follow the directions indicated by the arrows in the drawings.

First Embodiments

(1) Overall Outline

(26) An outline of an air conditioner **100** including an air conditioning indoor unit **30** according to one or more embodiments will be described with reference to FIG. **1**. FIG. **1** is a schematic configuration diagram of the air conditioner **100**.
(27) The air conditioner **100** performs a vapor compression refrigeration cycle to cool or heat an air conditioning target space R. The air conditioning target space R is, for example, a room of an office or a house. In one or more embodiments, the air conditioner **100** is capable of both cooling and heating the air conditioning target space R. However, the air conditioner of the present disclosure is not limited to an air conditioner capable of both cooling and heating, and may be, for example, a device capable of only cooling.
(28) The air conditioner **100** mainly includes an air conditioning heat source unit **10**, an air conditioning indoor unit **30**, a gas-refrigerant connection pipe GP and a liquid-refrigerant connection pipe LP that connect the air conditioning heat source unit **10** and the air conditioning indoor unit **30**.
(29) In one or more embodiments, the air conditioner **100** includes three air conditioning indoor units **30**. The number of the air conditioning indoor units **30** is not limited to three, but may be one, two, or four or more.
(30) The gas-refrigerant connection pipe GP and the liquid-refrigerant connection pipe LP are laid

at an installation site of the air conditioner **100**. The pipe diameters and pipe lengths of the gas-refrigerant connection pipe GP and the liquid-refrigerant connection pipe LP are selected according to design specifications and installation environments.
(31) In the air conditioner **100**, the air conditioning heat source unit **10** and the air conditioning indoor unit **30** are connected by the gas-refrigerant connection pipe GP and the liquid-refrigerant connection pipe LP to form a refrigerant circuit C. The refrigerant circuit C includes a compressor **12**, a heat-source heat exchanger **16**, and a first expansion valve **18** of the air conditioning heat source unit **10**, and a utilization heat exchanger **32** and a second expansion valve **34** of each air conditioning indoor unit **30**. The refrigerant circuit C includes a first shutoff valve **52** and a second shutoff valve **54** of each air conditioning indoor unit **30**.
(32) Although not limited, a flammable refrigerant is sealed in the refrigerant circuit C. Examples of the flammable refrigerant include refrigerants categorized in Class 3 (higher flammability), Class 2 (lower flammability), and Subclass 2L (slight flammability) in the standards according to ASHRAE 34 Designation and safety classification of refrigerant in the U.S.A. or the standards according to ISO 817 Refrigerants—Designation and safety classification. Examples of the refrigerant to be used herein may include, but not limited to, R1234yf, R1234ze(E), R516A, R445A, R444A, R454C, R444B, R454A, R455A, R457A, R459B, R452B, R454B, R447B, R32, R447A, R446A, and R459A. One or more embodiments adopt R32 as the refrigerant used therein. The air conditioning indoor unit and the air conditioner of the present disclosure are also useful when the refrigerant is not flammable. The air conditioning heat source unit **10** and the air conditioning indoor unit **30** will be described in detail below.
(2) Detailed Configuration
(33) (2-1) Air Conditioning Heat Source Unit
(34) The air conditioning heat source unit **10** will be described with reference to FIG. **1**.
(35) The air conditioning heat source unit **10** is installed, for example, on a rooftop of a building in which the air conditioner **100** is installed, in a machine room of the building, or around the building or the like.
(36) The air conditioning heat source unit **10** mainly includes a compressor **12**, a flow direction switching mechanism **14**, a heat-source heat exchanger **16**, a first expansion valve **18**, a first fan **20**, a first control unit **22**, a first stop valve **13***a*, and a second stop valve **13***b* (see FIG. **1**).
(37) The air conditioning heat source unit **10** includes, as refrigerant pipes, a suction pipe **11***a*, a discharge pipe **11***b*, a first gas refrigerant pipe **11***c*, a liquid refrigerant pipe **11***d*, and a second gas refrigerant pipe **11***e* (see FIG. **1**). The suction pipe **11***a* connects the flow direction switching mechanism **14** and a suction side of the compressor **12**. The discharge pipe **11***b* connects a discharge side of the compressor **12** and the flow direction switching mechanism **14**. The first gas refrigerant pipe **11***c* connects the flow direction switching mechanism **14** and a gas side end of the heat-source heat exchanger **16**. The liquid refrigerant pipe **11***d* connects a liquid side end of the heat-source heat exchanger **16** and the liquid-refrigerant connection pipe LP. The first stop valve **13***a* is provided at a connection portion between the liquid refrigerant pipe **11***d* and the liquid-refrigerant connection pipe LP. The first expansion valve **18** is provided in the liquid refrigerant pipe **11***d*. The second gas refrigerant pipe **11***e* connects the flow direction switching mechanism **14** and the gas-refrigerant connection pipe GP. The second stop valve **13***b* is provided at a connection portion between the second gas refrigerant pipe **11***e* and the gas-refrigerant connection pipe GP.
(38) (2-1-1) Compressor
(39) The compressor **12** sucks and compresses a low-pressure gas refrigerant in the refrigeration cycle and discharges a high-pressure gas refrigerant in the refrigeration cycle. The compressor **12** is, for example, an inverter control-type compressor. However, the compressor **12** may be a constant-speed compressor.
(40) (2-1-2) Flow Direction Switching Mechanism
(41) The flow direction switching mechanism **14** is a mechanism that switches the flow direction of

the refrigerant in the refrigerant circuit C according to the operation mode (cooling operation mode/heating operation mode) of the air conditioner **100**. The flow direction switching mechanism **14** is a four-way switching valve.
(42) In the cooling operation mode, the flow direction switching mechanism **14** switches the flow direction of the refrigerant in the refrigerant circuit C such that the refrigerant discharged from the compressor **12** is sent to the heat-source heat exchanger **16**. Specifically, in the cooling operation mode, the flow direction switching mechanism **14** causes the suction pipe **11***a* to communicate with the second gas refrigerant pipe **11***e* and causes the discharge pipe **11***b* to communicate with the first gas refrigerant pipe **11***c* (see the solid line in FIG. **1**). In the cooling operation mode, the heat-source heat exchanger **16** functions as a condenser, and the utilization heat exchanger **32** functions as an evaporator.
(43) In the heating operation mode, the flow direction switching mechanism **14** switches the flow direction of the refrigerant in the refrigerant circuit C such that the refrigerant discharged from the compressor **12** is sent to the utilization heat exchanger **32**. Specifically, in the heating operation mode, the flow direction switching mechanism **14** causes the suction pipe **11***a* to communicate with the first gas refrigerant pipe **11***c* and causes the discharge pipe **11***b* to communicate with the second gas refrigerant pipe **11***e* (see a broken line in FIG. **1**). In the heating operation mode, the heat-source heat exchanger **16** functions as an evaporator, and the utilization heat exchanger **32** functions as a condenser.
(44) The flow direction switching mechanism **14** may be realized without using a four-way switching valve. For example, the flow direction switching mechanism **14** may be configured by combining a plurality of electromagnetic valves and pipes so as to realize switching of the refrigerant flow direction as described above.
(45) (2-1-3) Heat-Source Heat Exchanger
(46) The heat-source heat exchanger **16** functions as a condenser of the refrigerant during the cooling operation, and functions as an evaporator of the refrigerant during the heating operation. Although not limited, the heat-source heat exchanger **16** is, for example, a fin-and-tube heat exchanger having a plurality of heat transfer tubes and a plurality of heat transfer fins.
(47) (2-1-4) First Expansion Valve
(48) The first expansion valve **18** is a mechanism that decompresses the refrigerant and adjusts the flow rate of the refrigerant. In one or more embodiments, the first expansion valve **18** is an electronic expansion valve whose opening degree is adjustable. The opening degree of the first expansion valve **18** is appropriately adjusted according to the operation situation. The first expansion valve **18** is not limited to the electronic expansion valve, and may be another type of expansion valve such as an automatic temperature expansion valve.
(49) (2-1-5) First Fan
(50) The first fan **20** is a blower that generates an air flow that flows into the air conditioning heat source unit **10** from the outside of the air conditioning heat source unit **10**, passes through the heat-source heat exchanger **16**, and then flows out to the outside of the air conditioning heat source unit **10**. The first fan **20** is, for example, an inverter control-type fan. However, the first fan **20** may be a constant-speed fan.
(51) (2-1-6) First Stop Valve and Second Stop Valve
(52) The first stop valve **13***a* is a valve disposed at a connection portion between the liquid refrigerant pipe **11***d* and the liquid-refrigerant connection pipe LP. The second stop valve **13***b* is a valve disposed at a connection portion between the second gas refrigerant pipe **11***e* and the gas-refrigerant connection pipe GP. The first stop valve **13***a* and the second stop valve **13***b* are manual valves. The first stop valve **13***a* and the second stop valve **13***b* are opened when the air conditioner **100** is used.
(53) (2-1-7) First Control Unit
(54) The first control unit **22** controls operations of various devices of the air conditioning heat

source unit **10**. The first control unit **22** mainly includes a microcontroller unit (MCU), various electric circuits, and electronic circuits (not shown). The MCU includes a CPU, a memory, an I/O interface, and the like. The memory of the MCU stores various programs to be executed by the CPU of the MCU. Note that the various functions of the first control unit **22** need not be implemented by software, and may be implemented by hardware or may be implemented by cooperation of hardware and software.
(55) The first control unit **22** is electrically connected to various devices of the air conditioning heat source unit **10** including the compressor **12**, the flow direction switching mechanism **14**, the first expansion valve **18**, and the first fan **20** (see FIG. **1**). The first control unit **22** is electrically connected to various sensors (not shown) provided in the air conditioning heat source unit **10**. Although not limited, the sensors provided in the air conditioning heat source unit **10** include a temperature sensor and a pressure sensor provided in the discharge pipe **11***b* and the suction pipe **11***a*, a temperature sensor provided in the heat-source heat exchanger **16** and the liquid refrigerant pipe **11***d*, a temperature sensor that measures the temperature of the heat source air, and the like. However, the air conditioning heat source unit **10** does not need to include all these sensors.
(56) The first control unit **22** is connected to the second control unit **38** of the air conditioning indoor unit **30** via a communication line. The first control unit **22** and the second control unit **38** exchange various signals via a communication line. The first control unit **22** and the second control unit **38** cooperate to function as a controller **90** that controls the operation of the air conditioner **100**. The function of the controller **90** will be described later.
(57) (2-2) Air Conditioning Indoor Unit
(58) The air conditioning indoor unit **30** will be described with reference to FIGS. **2** to **5** in addition to FIG. **1**.
(59) FIG. **2** is an external perspective view of the air conditioning indoor unit **30**. FIG. **3** is a schematic cross-sectional view of the air conditioning indoor unit **30** attached to the ceiling CL, taken along line III-III in FIG. **2**. FIG. **4** is a schematic bottom view of the air conditioning indoor unit **30**. FIG. **4** illustrates the air conditioning indoor unit **30** with a decorative plate **46** and a bottom plate **48** removed. FIG. **5** is a schematic perspective view around the utilization heat exchanger **32** for explaining the first space S**1** in which the first shutoff valve **52** and the second shutoff valve **54** are disposed. In FIG. **5**, the casing **40**, the second expansion valve **34**, the second fan **36**, and the like are not shown from the viewpoint of visibility of the drawing.
(60) In one or more embodiments, the air conditioner **100** includes three air conditioning indoor units **30** having the same structure. The three air conditioning indoor units **30** may not necessarily be identical. For example, the air conditioning indoor unit **30** may have different capabilities.
(61) The air conditioning indoor unit **30** blows the air having exchanged heat with the refrigerant flowing through the use heat exchanger **32** to the air conditioning target space R. In one or more embodiments, the air conditioning indoor unit **30** is a ceiling-mounted type installed on the ceiling of the air conditioning target space R. In particular, the air conditioning indoor unit **30** according to one or more embodiments is a ceiling-embedded type air conditioning indoor unit. Examples of the ceiling-embedded type air conditioning indoor unit include a ceiling cassette-type air conditioning indoor unit in which at least a part of the air conditioning indoor unit is disposed in the attic space CS, and a duct connection type air conditioning indoor unit in which the entire air conditioning indoor unit is disposed in the attic space CS and a duct is connected. However, the type of the air conditioning indoor unit **30** is not limited to the ceiling-embedded type, and may be a ceiling-suspended type. The air conditioning indoor unit **30** may be of a type other than a ceiling-mounted type such as a wall-mounted type or a floor type.
(62) As shown in FIGS. **1** and **3**, the air conditioning indoor unit **30** mainly includes a casing **40**, a utilization heat exchanger **32**, a second expansion valve **34**, a second fan **36**, a first shutoff valve **52**, a second shutoff valve **54**, a refrigerant detector **56**, and a second control unit **38**.
(63) The air conditioning indoor unit **30** also includes, as refrigerant pipes, a liquid refrigerant pipe

37*a* and a gas refrigerant pipe **37***b* connected to the utilization heat exchanger **32** (see FIG. **1**). The liquid refrigerant pipe **37***a* connects the liquid-refrigerant connection pipe LP and the liquid side of the utilization heat exchanger **32**. The liquid refrigerant pipe **37***a* is provided with a first shutoff valve **52**. A second expansion valve **34** is provided between the first shutoff valve **52** and the utilization heat exchanger **32** in the liquid refrigerant pipe **37***a*. The gas refrigerant pipe **37***b* connects the gas-refrigerant connection pipe GP and the gas side of the utilization heat exchanger **32**. The gas refrigerant pipe **37***b* is provided with a second shutoff valve **54**.

(64) (2-2-1) Casing

(65) The casing **40** accommodates various devices of the air conditioning indoor unit **30**. The various devices accommodated in the casing **40** mainly include the utilization heat exchanger **32**, the second expansion valve **34**, the second fan **36**, the first shutoff valve **52**, and the second shutoff valve **54** (see FIGS. **3** and **4**).

(66) As illustrated in FIG. **3**, the casing **40** is inserted into an opening formed in the ceiling CL of the target space, and is installed in the attic space CS formed between the ceiling CL and the floor surface of the upper floor or between the ceiling CL and the roof. The casing **40** includes a top panel **42***a*, side walls **42***b*, a bottom plate **48**, and a decorative plate **46** (see FIGS. **2** and **3**).

(67) The top panel **42***a* is a member constituting a top surface portion of the casing **40**. In a plan view, the top panel **42***a* has a substantially quadrangular shape (see FIG. **4**).

(68) The side wall **42***b* is a member constituting a side surface portion of the casing **40**. The side wall **42***b* extends downward from the top panel **42***a*. The side wall **42***b* has a substantially quadrangular prism shape corresponding to the shape of the top panel **42***a*. Although the material is not limited, the side wall **42***b* and the top panel **42***a* are made of sheet metal, for example. The side wall **42***b* and the top panel **42***a* are integrally formed, and, as a whole, have a substantially quadrangular box shape in a plan view with a lower surface opened. An opening **44** through which the liquid refrigerant pipe **37***a* and the gas refrigerant pipe **37***b* connected to the utilization heat exchanger **32** are inserted is formed in the side wall **42***b* (see FIG. **4**). The liquid-refrigerant connection pipe LP is connected to an end of the liquid refrigerant pipe **37***a* disposed outside the casing **40**. The gas-refrigerant connection pipe GP is connected to an end of the gas refrigerant pipe **37***b* disposed outside the casing **40**. For example, a flare nut is used for connection between the liquid refrigerant pipe **37***a* and the liquid-refrigerant connection pipe LP and connection between the gas refrigerant pipe **37***b* and the gas-refrigerant connection pipe GP. The connection between the liquid refrigerant pipe **37***a* and the liquid-refrigerant connection pipe LP and the connection between the gas refrigerant pipe **37***b* and the gas-refrigerant connection pipe GP may be performed by welding or brazing.

(69) The bottom plate **48** is a member constituting a bottom surface portion of the casing **40**. The material of the bottom plate **48** is not limited, but the bottom plate is made of styrene foam. A part of the bottom plate **48** functions as a drain pan. Specifically, a first portion **48***a* of the bottom plate **48** which is disposed below the utilization heat exchanger **32** and has a groove recessed downward for receiving the condensed water functions as a drain pan. As illustrated in FIGS. **3** and **4** (indicated by a two-dot chain line in FIG. **4**), a suction opening **481** having a substantially circular shape in a plan view is formed at the center of the bottom plate **48**. A bell mouth **50** is disposed at the suction opening **481**. As shown in FIGS. **3** and **4** (indicated by a two-dot chain line in FIG. **4**), a plurality of blow-out openings **482** are formed around the suction opening **481** of the bottom plate **48**. As shown in FIGS. **2** and **3**, the decorative plate **46** is attached to the lower surface side of the bottom plate **48**.

(70) The decorative panel **46** is a plate-shaped member exposed to the air conditioning target space R. The decorative plate **46** has a substantially quadrangular shape in a plan view. The decorative plate **46** is installed by being fitted into an opening of the ceiling CL (see FIG. **3**). The decorative plate **46** includes an air suction port **46***a* and a plurality of blow-out ports **46***b*. The suction port **46***a* is formed in a substantially quadrangular shape in a central portion of the decorative plate **46** at a

position partially overlapping the suction opening **481** of the bottom plate **48** in a plan view. The plurality of blow-out ports **46***b* are formed around the suction port **46***a* of the decorative plate **46** so as to surround the suction port **46***a*. Each of the blow-out ports **46***b* is disposed at a position corresponding to the blow-out opening **482** of the bottom plate **48**. When the second fan **36** is operated, air sucked from the suction port **46***a* flows into the casing **40** through the suction opening **481**. The air that has flowed into the casing **40** and passed through the utilization heat exchanger **32** is blown out from the blow-out opening **482** and is blown out into the air conditioning target space R from the blow-out port **46***b* corresponding to the blow-out opening **482** (see FIG. **3**).
(71) The arrangement of devices, components, and spaces in the casing **40** will be described.
(72) As illustrated in FIG. **4**, the second fan **36** is disposed at the center of the casing **40** in a plan view. As illustrated in FIG. **3**, a bell mouth **50** is provided below the second fan **36**. As illustrated in FIG. **4**, in a plan view, a utilization heat exchanger **32** is provided around the second fan **36** so as to surround the second fan **36**. As described above, the groove recessed downward is formed in the first portion **48***a* of the bottom plate **48** disposed below the utilization heat exchanger **32**. The first portion **48***a* of the bottom plate **48** functions as a drain pan that receives condensed water generated in the utilization heat exchanger **32** (see FIG. **3**).
(73) In a plan view, as shown in FIG. **4**, a first space S**1** separated from the second space S**2** by a partition wall **60** is formed at one of the corners of the casing **40**. The second space S**2** communicates with the air conditioning target space R through the suction port **46***a* and the suction opening **481**, and the blow-out opening **482** and the blow-out port **46***b*. The second space S**2** includes an air flow path through which air flows from the suction port **46***a* to the blow-out port **46***b* via the utilization heat exchanger **32** during operation of the second fan **36**. The presence of the partition wall **60** suppresses the flow of air between the first space S**1** and the second space S**2**. Therefore, even if the refrigerant leaks in the first space S**1**, the inflow of the refrigerant from the first space S**1** to the second space S**2** is suppressed. Furthermore, even if the refrigerant leaks in the first space S**1**, the inflow of the refrigerant from the first space S**1** to the air conditioning target space R via the second space S**2** is suppressed.
(74) The air may not flow between the first space S**1** and the second space S**2**. Here, “air may not flow between the first space S**1** and the second space S**2**” means that there is substantially no air flow, and the first space S**1** and the second space S**2** may not be sealed in an airtight state.
(75) As shown in FIGS. **4** and **5**, the first space S**1** is a space formed such that the upper side is surrounded by the top panel **42***a* of the casing **40**, the lateral sides are surrounded by the side wall **42***b* and the partition wall **60** of the casing **40**, and the lower side is surrounded by the bottom plate **48**. So as not to communicate the first space S**1** and the second space S**2** each other, the portion of the bottom plate **48** surrounding the first space S**1** does not include the first portion **48***a* functioning as a drain pan.
(76) The partition wall **60** is a plate-like member here. The partition wall **60** is attached to, for example, a tube plate **32***a* of the utilization heat exchanger **32**. The tube plate **32***a* is a member for fixing a plurality of heat transfer tubes (not shown) of the utilization heat exchanger **32**, and is provided at both ends of the heat transfer tubes. The partition wall **60** includes a first member **62** connecting the two tube plates **32***a* of the utilization heat exchanger **32** and a second member **64** extending from the tube plates **32***a* toward the side wall **42***b* of the casing **40**. The second member **64** may come into contact with the side wall **42***b* directly or indirectly via another member. Since the partition wall **60** and the side wall **42***b* are in direct or indirect contact with each other, the flow of air between the first space S**1** and the second space S**2** is easily suppressed. The partition wall **60** may be in contact with the top panel **42***a* and the bottom plate **48** of the casing **40** directly or indirectly via another member. Since the partition wall **60** is in direct or indirect contact with the top panel **42***a* and the bottom plate **48**, the flow of air between the first space S**1** and the second space S**2** is easily suppressed. Note that a sealing material may be appropriately used in order to suppress the flow of air between the first space S**1** and the second space S**2**. Note that the structure

for forming the first space S**1** described here is merely an example, and the first space S**1** may be formed in another manner. For example, the upper side of the first space S**1** may be surrounded by a member separate from the casing **40** instead of the top panel **42***a* of the casing **40**. The lower side of the first space S**1** may be surrounded by a member that is not formed integrally with the bottom plate **48** of the casing **40**.
(77) An opening **44** through which the liquid refrigerant pipes **37***a* and the gas refrigerant pipes **37***b* pass is formed in the side wall **42***b* of the casing **40** that forms the first space S**1**, in other words, that surrounds the first space S**1**. The first space S**1** and the attic space CS in which the casing **40** is installed communicate with each other via the opening **44**. The first space S**1** and the attic space CS may communicate with each other via the opening **44**, but gaps between the liquid refrigerant pipe **37***a* and the gas refrigerant pipe **37***b* and the opening **44** may be closed by a sealing material or the like.
(78) The second expansion valve **34**, the first shutoff valve **52**, and the second shutoff valve **54** are disposed in the first space S**1**. The second expansion valve **34**, the first shutoff valve **52**, and the second shutoff valve **54** are disposed, for example, at a lower part of the first space S**1**. However, the present disclosure is not limited to this, and the position in which the second expansion valve **34**, the first shutoff valve **52**, and the second shutoff valve **54** are disposed in the first space S**1** may be appropriately determined. The position of the first space S**1** described here is an example, and the first space S**1** may be formed at a place other than the corner of the casing **40** in a plan view.
(79) (2-2-2) Indoor Heat Exchanger
(80) The utilization heat exchanger **32** is an example of the heat exchanger. In the utilization heat exchanger **32**, heat is exchanged between the refrigerant flowing through the utilization heat exchanger **32** and air.
(81) The type of the utilization heat exchanger **32** is not limited, and is, for example, a fin-and-tube heat exchanger having a plurality of heat transfer tubes and a plurality of heat transfer fins.
(82) Although the shape and structure are not limited, the utilization heat exchanger **32** illustrated in FIGS. **4** and **5** has a plurality of rows of heat exchange units **33** in which a plurality of heat transfer tubes are vertically arranged and stacked. Here, the utilization heat exchanger **32** includes two rows of heat exchange units **33**. The heat exchange units **33** of the utilization heat exchanger **32** are arranged along the flow direction of the air generated by the second fan **36**. Tube plates **32***a* for fixing the heat transfer tubes are provided at both ends of the heat exchange unit **33**. As shown in FIG. **4**, the heat exchange unit **33** of the utilization heat exchanger **32** is bent by about 90 degrees at three points in a plan view, and is disposed in a substantially quadrangular shape. The utilization heat exchanger **32** is disposed so as to surround the suction port **46***a* and to be surrounded by the blow-out port **46***b* in a plan view. The utilization heat exchanger **32** is disposed so as to surround the periphery of the second fan **36**.
(83) As illustrated in FIG. **1**, the liquid refrigerant pipe **37***a* is connected to one end of the utilization heat exchanger **32**. As illustrated in FIG. **1**, the gas refrigerant pipe **37***b* is connected to the other end of the utilization heat exchanger **32**. Specifically, in one or more embodiments, the liquid refrigerant pipe **37***a* is connected to a first header **32***b* of the utilization heat exchanger **32**. The gas refrigerant pipe **37***b* is connected to a second header **32***c* of the utilization heat exchanger **32**.
(84) During the cooling operation, the refrigerant flows from the liquid refrigerant pipe **37***a* into the utilization heat exchanger **32**, and the refrigerant having exchanged heat with air in the heat exchange unit **33** of the utilization heat exchanger **32** flows out from the gas refrigerant pipe **37***b*. During the heating operation, the refrigerant flows from the gas refrigerant pipe **37***b* into the utilization heat exchanger **32**, and the refrigerant having exchanged heat with air in the heat exchange unit **33** of the utilization heat exchanger **32** flows out from the liquid refrigerant pipe **37***a*.
(85) (2-2-3) Second Expansion Valve
(86) The second expansion valve **34** is a mechanism that decompresses the refrigerant and adjusts

the flow rate of the refrigerant. In one or more embodiments, the second expansion valve **34** is an electronic expansion valve whose opening degree is adjustable. The opening degree of the second expansion valve **34** is appropriately adjusted according to the operation situation. The second expansion valve **34** is not limited to the electronic expansion valve, and may be another type of expansion valve such as an automatic temperature expansion valve.
(87) (2-2-4) Second Fan
(88) The second fan **36** is a blower that supplies air to the utilization heat exchanger **32**. The second fan **36** is, for example, a centrifugal fan such as a turbo fan or a sirocco fan. The second fan **36** is, for example, but not limited to, an inverter control-type fan.
(89) When the second fan **36** is operated, the air in the air conditioning target space R flows into the casing **40** of the air conditioning indoor unit **30** from the suction port **46***a* of the decorative plate **46**, passes through the bell mouth **50**, is sucked into the second fan **36**, and is blown out in four directions from the second fan **36**. The air blown out by the second fan **36** passes through the utilization heat exchanger **32** toward the blow-out port **46***b*, and is blown out from the blow-out port **46***b* into the air conditioning target space R. At least a portion of the second space S**2** functions as a flow path of air through which air flows in the above-described manner during operation of the second fan **36**. Since the partition wall **60** is present, the air blown out by the second fan **36** hardly flows into the first space S**1**.
(90) (2-2-5) First Shutoff Valve and Second Shutoff Valve
(91) The first shutoff valve **52** and the second shutoff valve **54** suppress the leakage of refrigerant into the air conditioning target space R when the leakage of refrigerant from the refrigerant circuit C. The first shutoff valve **52** and the second shutoff valve **54** are, for example, electromagnetic valves capable of switching between a closed state (fully closed) and an open state (fully open). However, the types of the first shutoff valve **52** and the second shutoff valve **54** are not limited to the electromagnetic valve, and may be, for example, an electric valve.
(92) The first shutoff valve **52** and the second shutoff valve **54** are opened in a normal state (when the refrigerant detector **56** does not detect the leakage of refrigerant). When the refrigerant detector **56** of the air conditioning indoor unit **30** detects the leakage of refrigerant, the first shutoff valve **52** and the second shutoff valve **54** of the air conditioning indoor unit **30** are closed. When the first shutoff valve **52** and the second shutoff valve **54** are closed while the refrigerant leaks from the air conditioning indoor unit **30**, the inflow of the refrigerant into the air conditioning indoor unit **30** from the air conditioning heat source unit **10**, the pipe connecting the air conditioning heat source unit **10** and the first shutoff valve **52**, or the pipe connecting the air conditioning heat source unit **10** and the second shutoff valve **54** is suppressed.
(93) (2-2-6) Refrigerant Detector
(94) The refrigerant detector **56** is a sensor that detects the leakage of refrigerant at the air conditioning indoor unit **30**.
(95) The refrigerant detector **56** is provided in the casing **40** of the air conditioning indoor unit **30**, for example. As illustrated in FIG. **3**, the refrigerant detector **56** is attached to the bottom surface of the bottom plate **48** disposed below the utilization heat exchanger **32**. The refrigerant detector **56** may be attached to a place other than the bottom plate **48**, for example, a bottom surface of a member connecting the bell mouth **50** and the bottom plate **48**, a bottom surface of the bell mouth **50**, an inner surface of the top panel **42***a* or the side wall **42***b*, or the like. The refrigerant detector **56** may be disposed outside the casing **40** of the air conditioning indoor unit **30**. A plurality of refrigerant detectors **56** may be provided.
(96) The refrigerant detector **56** is, for example, a semiconductor-type sensor. The semiconductor refrigerant detector **56** includes a semiconductor-type detection element (not shown). The semiconductor detector element has electric conductivity that changes depending on whether it is in a case where no refrigerant gas exists therearound and in a case where refrigerant gas exists therearound. When the refrigerant gas exists around the semiconductor-type detection element, the

refrigerant detector **56** outputs a relatively large current as a detection signal. On the other hand, when the refrigerant gas does not exist around the semiconductor-type detection element, the refrigerant detector **56** outputs a relatively small current as a detection signal.
(97) The type of the refrigerant detector **56** is not limited to the semiconductor type, and may be any sensor capable of detecting refrigerant gas. For example, the refrigerant detector **56** may be an infrared sensor that outputs a detection signal according to a detection result of the refrigerant.
(98) (2-2-7) Second Control Unit
(99) The second control unit **38** controls operations of various devices of the air conditioning indoor unit **30**. The second control unit **38** includes a microcontroller unit (MCU), various electric circuits, and electronic circuits (not shown). The MCU includes a CPU, a memory, an I/O interface, and the like. The memory of the MCU stores various programs to be executed by the CPU of the MCU. Note that the various functions of the second control unit **38** need not be implemented by software, and may be implemented by hardware or may be implemented by cooperation of hardware and software.
(100) The second control unit **38** is electrically connected to various devices of the air conditioning indoor unit **30** including the second expansion valve **34**, the second fan **36**, the first shutoff valve **52**, and the second shutoff valve **54** (see FIG. **1**). The second control unit **38** is electrically connected to the refrigerant detector **56**. The second control unit **38** is electrically connected to a sensor (not shown) provided in the air conditioning indoor unit **30**. The sensor (not shown) includes, but is not limited to, a temperature sensor provided in the utilization heat exchanger **32** and the liquid refrigerant pipe **37***a*, a temperature sensor that measures the temperature of the air conditioning target space R, and the like.
(101) The second control unit **38** is connected to the first control unit **22** of the air conditioning heat source unit **10** via a communication line. The second control unit **38** is communicably connected to a remote control unit for operating the air conditioner **100** (not shown) via a communication line. The first control unit **22** and the second control unit **38** cooperate to function as a controller **90** that controls the operation of the air conditioner **100**.
(102) The function of the controller **90** will be described. Some or all of various functions of the controller **90** described below may be executed by a control device provided separately from the first control unit **22** and the second control unit **38**.
(103) The controller **90** controls the operation of the flow direction switching mechanism **14** such that the heat-source heat exchanger **16** functions as a condenser of the refrigerant and the utilization heat exchanger **32** functions as an evaporator of the refrigerant during the cooling operation. The controller **90** controls the operation of the flow direction switching mechanism **14** such that the heat-source heat exchanger **16** functions as an evaporator of the refrigerant and the utilization heat exchanger **32** functions as a condenser of the refrigerant during the heating operation. The controller **90** operates the compressor **12**, the first fan **20**, and the second fan **36** during the cooling operation and the heating operation. During the cooling operation and the heating operation, the controller **90** adjusts the numbers of rotations of the motors of the compressor **12**, the first fan **20**, and the second fan **36** and the opening degrees of the first expansion valve **18** and the second expansion valve **34** based on the measurement values of the various temperature sensors and the pressure sensors, the set temperatures, and the like. Since various control modes are generally known for the control of the operations of the various devices of the air conditioner **100** during the cooling operation and the heating operation, the description thereof will be omitted here to avoid complication of the description.
(104) In addition to the control of the normal operation of the air conditioner **100**, the controller **90** performs the following control when the refrigerant is detected by the refrigerant detector **56** of any of the air conditioning indoor units **30**. The case where the refrigerant is detected by the refrigerant detector **56** means a case where the value of the current output as the detection signal by the refrigerant detector **56** is larger than a predetermined threshold.

(105) When the refrigerant is detected by the refrigerant detector **56** of any of the air conditioning indoor units **30**, the controller **90** closes the first shutoff valve **52** and the second shutoff valve **54** of the air conditioning indoor unit **30**. When the refrigerant is detected by the refrigerant detector **56** of any of the air conditioning indoor units **30**, the controller **90** may notify the leakage of refrigerant using an alarm (not shown) in addition to the control to close the first shutoff valve **52** and the second shutoff valve **54** in the air conditioning indoor unit **30** in which the refrigerant is detected. When the refrigerant is detected by the refrigerant detector **56** of any of the air conditioning indoor units **30**, the controller **90** may stop the operation of the entire air conditioner **100** by stopping the operation of the compressor **12** of the air conditioning heat source unit **10** in addition to the control to close the first shutoff valve **52** and the second shutoff valve **54** in the air conditioning indoor unit **30** in which the refrigerant is detected.
(3) Features
(106) (3-1)
(107) The air conditioning indoor unit **30** according to the above-described embodiments blows air that has exchanged heat with the refrigerant flowing through the utilization heat exchanger **32** into the air conditioning target space R. The utilization heat exchanger **32** is an example of a heat exchanger. The air conditioning indoor unit **30** includes a liquid refrigerant pipe **37***a* and a gas refrigerant pipe **37***b* connected to the utilization heat exchanger **32**, a casing **40**, a first shutoff valve **52** and a second shutoff valve **54**, and a partition wall **60**. The casing **40** accommodates the utilization heat exchanger **32**. An opening communicating with the air conditioning target space R is formed in the casing **40**. The opening includes a suction port **46***a* and a suction opening **481** for sucking air into the casing **40**. The opening includes a blow-out port **46***b* and a blow-out opening **482** through which air is blown out of the casing **40**. The first shutoff valve **52** and the second shutoff valve **54** are disposed in the first space S**1** in the casing **40**. The first shutoff valve **52** is disposed in the liquid refrigerant pipe **37***a*. The second shutoff valve **54** is disposed in the gas refrigerant pipe **37***b*. The partition wall **60** separates the first space S**1** and the second space S**2**. The second space S**2** is a space in the casing **40** and communicates with the air conditioning target space R via an opening.
(108) In the air conditioning indoor unit **30**, the first shutoff valve **52** and the second shutoff valve **54** are disposed in the casing **40** of the air conditioning indoor unit **30**. Therefore, the amount of work for installing the air conditioner **100** on site can be reduced, as compared with the case where providing the shutoff valves to the refrigerant connection pipes LP and GP laid on site during the air conditioning heat source unit **10** and the air conditioning indoor unit **30** are connected.
(109) In the air conditioning indoor unit **30**, the first shutoff valve **52** and the second shutoff valve **54** are disposed in the first space S**1** separated from the second space S**2** communicating with the air conditioning target space R by the partition wall **60**. In other words, the first shutoff valve **52** and the second shutoff valve **54** are disposed in the first space S**1** where the flow of air to the air conditioning target space R is suppressed. Therefore, in the air conditioning indoor unit **30**, even if the refrigerant leaks around the first shutoff valve **52** and the second shutoff valve **54**, the outflow of the refrigerant into the air conditioning target space R can be suppressed. Therefore, safety is high even when a flammable refrigerant is used.
(110) In the air conditioning indoor unit **30**, the first shutoff valve **52** and the second shutoff valve **54** are disposed in the casing **40** of the air conditioning indoor unit **30**. Therefore, the amount of the refrigerant flowing out of the refrigerant leaking location when the leakage of refrigerant occurs in the air conditioning indoor unit **30** can be reduced as compared with the case where the shutoff valve is installed at a position of the connection pipes LP and GP distant from the air conditioning indoor unit **30**. The reason why the amount of the refrigerant flowing out of the refrigerant leaking location may increase when the shutoff valve is installed at a position in the connection pipes LP and GP distant from the air conditioning indoor unit **30** is that the refrigerant in the connection pipes LP and GP between the shutoff valve and the air conditioning indoor unit **30** may also flow

out of the refrigerant leaking location of the air conditioning indoor unit **30**.

(111) Further, when the air conditioning indoor unit **30** is used, it is not necessary to secure a space for installing the shutoff valve outside the casing **40** of the air conditioning indoor unit **30**, and construction is easy.

(112) (3-2)

(113) In the air conditioning indoor unit **30** according to the above-described embodiments, the first space S**1** communicates with the attic space CS.

(114) Therefore, even if the refrigerant leaks around the first shutoff valve **52** and the second shutoff valve **54**, the refrigerant flows into the attic space CS that does not directly communicate with the air conditioning target space R. Accordingly, the air conditioning indoor unit **30** suppresses inflow of the refrigerant into the air conditioning target space R and thus achieves high safety.

(115) (3-3)

(116) In the air conditioner **100** of the above-described embodiments, the first shutoff valve **52** and the second shutoff valve **54** are disposed inside the casing **40** of the air conditioning indoor unit **30**. Therefore, the amount of work for installing the air conditioner **100** on site can be reduced as compared with the case where providing the shutoff valves to the refrigerant connection pipes LP and GP laid on site during the air conditioning heat source unit **10** and the air conditioning indoor unit **30** are connected.

(4) Modifications

(117) The above-mentioned embodiments may be appropriately modified as described in the following modifications. Each modification may be applied in combination with another modification as long as no contradiction occurs.

(118) (4-1) Modification 1A

(119) The first shutoff valve **52** and the second shutoff valve **54** of the above-described embodiments are dedicated valves for preventing refrigerant leakage. However, valves used for purposes other than the purpose of preventing refrigerant leakage may be used as the first shutoff valve **52** and the second shutoff valve **54** for preventing refrigerant leakage.

(120) For example, like the air conditioning indoor unit **30***a* in FIG. **6**, the first shutoff valve **52** of the above-described embodiments may be omitted, and the electronic expansion valve as the second expansion valve **34** disposed in the first space S**1** may also be used as the first shutoff valve. Specifically, when the refrigerant detector **56** of any of the air conditioning indoor units **30** detects the leakage of refrigerant, the controller **90** may control to close (fully close) the second expansion valve **34** as the first shutoff valve and the second shutoff valve **54** of that air conditioning indoor unit **30**.

(121) The air conditioner **100***a* illustrated in FIG. **6** is similar to the air conditioner **100** of the above-described embodiments except that the second expansion valve **34** is also used as the first shutoff valve, and thus detailed description thereof is omitted.

(122) (4-2) Modification 1B

(123) In the above-described embodiments, the first space S**1** communicates with the attic space CS. Alternatively, the first space S**1** may communicate with a space other than the attic space CS that does not directly communicate with the air conditioning target space R. For example, the first space S**1** may communicate with a space under the floor, a pipe space, or the like that does not communicate with the air conditioning target space R. When the refrigerant is flammable, the space communicating with the first space S**1** may be a space without an ignition source.

(124) (4-3) Modification 1C

(125) In the above-described embodiments, the ceiling cassette-type air conditioning indoor unit **30** in which a part (decorative plate **46**) of the casing **40** is exposed to the interior is described as a specific example of the air conditioning indoor unit of the present disclosure. However, the type of the air conditioning indoor unit **30** is not limited to the ceiling cassette type. The air conditioning indoor unit of the present disclosure may be, for example, a duct connection-type air conditioning

indoor unit, which is one of embedded ceiling types and in which the entire air conditioning indoor unit is disposed in the attic space CS and a duct communicating with the air conditioning target space R is connected to the air conditioning indoor unit.

(126) A specific example of the duct connection-type air conditioning indoor unit **130** in which the first shutoff valve **52** and the second shutoff valve **54** are disposed in the first space S**1** will be described with reference to FIG. **7**. FIG. **7** is a schematic plan view for explaining an internal structure and device arrangement of the air conditioning indoor unit **130**. In FIG. **7**, the top panel of the casing **140** of the air conditioning indoor unit **130** is not shown.

(127) The utilization heat exchanger **132**, the second fan **136**, the second expansion valve **34**, the first shutoff valve **52**, and the second shutoff valve **54** of the air conditioning indoor unit **130** are functionally identical to the utilization heat exchanger **32**, the second fan **36**, the second expansion valve **34**, the first shutoff valve **52**, and the second shutoff valve **54** of the above-described embodiments, respectively. Therefore, detailed description of the utilization heat exchanger **132**, the second fan **136**, the second expansion valve **34**, the first shutoff valve **52**, and the second shutoff valve **54** is omitted unless otherwise necessary in the description of the present disclosure.

(128) As illustrated in FIG. **7**, the air conditioning indoor unit **130** includes a casing **140** that accommodates the utilization heat exchanger **132**, the second fan **136**, the second expansion valve **34**, the first shutoff valve **52**, and the second shutoff valve **54**. In the air conditioning indoor unit **130**, the entire casing **140** is disposed in the attic space CS. In other words, the casing **140** is disposed at a position normally invisible from the air conditioning target space R.

(129) The casing **140** mainly includes a top panel (not shown), a side wall **142***b*, a bottom plate **142***c*, a partition plate **142***d*, and a first member **148**.

(130) Although the material is not limited, the top panel, the side wall **142***b*, the bottom plate **142***c*, and the partition plate **142***d* of the casing **140** are made of sheet metal, for example. Although the material is not limited, the first member **148** of the casing **140** is made of, for example, styrene foam.

(131) The top panel of the casing **140** is a member constituting a top surface portion of the casing **140**. The top panel of the casing **140** has a substantially quadrangular shape in a plan view.

(132) The side wall **142***b* is a member constituting a side surface portion of the casing **140**. The side wall **142***b* extends downward from the top panel of the casing **140**. The side wall **142***b* has a substantially quadrangular shape corresponding to the shape of the casing **140**.

(133) The side wall **142***b* is provided with an opening **144** through which the liquid refrigerant pipe **37***a* and the gas refrigerant pipe **37***b* connected to the utilization heat exchanger **132** are inserted. In FIG. **7**, an opening **144** through which the liquid refrigerant pipe **37***a* and the gas refrigerant pipe **37***b* connected to the utilization heat exchanger **132** are inserted is formed in the side wall **142***b* disposed on the left side of the casing **140** (see FIG. **7**). The liquid-refrigerant connection pipe LP is connected to an end of the liquid refrigerant pipe **37***a* disposed outside the casing **140**. The gas-refrigerant connection pipe GP is connected to an end of the gas refrigerant pipe **37***b* disposed outside the casing **140**. The connection between the liquid refrigerant pipe **37***a* and the liquid-refrigerant connection pipe LP and the connection between the gas refrigerant pipe **37***b* and the gas-refrigerant connection pipe GP are the same as those in the above-described embodiments, and thus the description thereof will be omitted.

(134) A suction opening **144***a* to which a suction duct ID for taking in air from air conditioning target space R is connected is formed in the side wall **142***b* disposed on the rear side of the casing **140**. The side wall **142***b* disposed on the front side of the casing **140** is provided with a blow-out opening **144***b* to which a blow-out duct OD that supplies air to the air conditioning target space R is connected. The space inside the casing **140** and the attic space CS do not communicate with each other through the suction opening **144***a* or the blow-out opening **144***b*. In other words, the air in the space in the attic space CS does not substantially flow into the casing **140** from the suction opening **144***a* or the blow-out opening **144***b*.

(135) The bottom plate **142***c* of the casing **140** is a member constituting a bottom surface portion of the casing **140**. In a plan view, the bottom plate **142***c* of the casing **140** has a substantially quadrangular shape.
(136) The partition plate **142***d* of the casing **140** is a member that partitions the inside of the casing **140** into a fan chamber in which the second fan **136** is mainly disposed and a heat exchange chamber in which the utilization heat exchanger **132** is mainly disposed. The partition plate **142***d* prevents air from flowing between a fan chamber (a space on the rear side of the partition plate **142***d* in FIG. **7**) and a heat exchange chamber (a space on the front side of the partition plate **142***d* in FIG. **7**). However, the partition plate **142***d* is formed with an opening **142***da* through which a blow-out portion **136***a* of the second fan **136** is inserted in order to dispose the blow-out portion **136***a* of the second fan **136** in the heat exchange chamber. When the second fan **136** is operated, the air sucked from the air conditioning target space R through the suction duct ID and the suction opening **144***a* is blown out from the blow-out portion **136***a* of the second fan **136** toward the utilization heat exchanger **132**. In other words, the air in the fan chamber does not directly flow into the heat exchange chamber, but flows into the heat exchange chamber via the second fan **136**. The air blown out of the second fan **136** exchanges heat with the refrigerant flowing through the utilization heat exchanger **132**, and blows out into the air conditioning target space R through the blow-out opening **144***b* and the blow-out duct OD (see arrows in FIG. **7**).
(137) The first member **148** is a member disposed in a space on the front side of the partition plate **142***d* in the casing **140**, above the bottom plate **142***c* of the casing **140**, and below the utilization heat exchanger **132**. In a portion of the first member **148** disposed below the utilization heat exchanger **132**, a recess (not shown) is formed so as to be recessed downward to receive condensed water generated in the utilization heat exchanger **132**. The recess of the first member **148** disposed below the utilization heat exchanger **132** functions as a drain pan.
(138) As illustrated in FIG. **7**, the first space S**1** is formed on the left side of the utilization heat exchanger **132**. The first space S**1** is separated from the second space S**2** by a partition wall **160**. The second space S**2** herein communicates with the air conditioning target space R via the suction opening **144***a* and the blow-out opening **144***b*. The second space S**2** includes a flow path of air through which air flows from the blow-out portion **136***a* of the second fan **136** to the blow-out opening **144***b* via the utilization heat exchanger **32** during operation of the second fan **136**. The presence of the partition wall **160** suppresses the flow of air between the first space S**1** and the second space S**2**. Therefore, even if the refrigerant leaks in the first space S**1**, the inflow of the refrigerant from the first space S**1** to the second space S**2** is suppressed. Furthermore, even if the refrigerant leaks in the first space S**1**, the inflow of the refrigerant from the first space S**1** to the air conditioning target space R via the second space S**2** is suppressed.
(139) The air may not flow between the first space S**1** and the second space S**2**. Here, “air may not flow between the first space S**1** and the second space S**2**” means that there is substantially no air flow, and the first space S**1** and the second space S**2** may not be sealed in an airtight state.
(140) The first space S**1** is a space formed such that the upper side is surrounded by a top panel (not shown) of the casing **140**, the lateral sides are surrounded by the side wall **142***b*, the partition wall **160**, and the partition plate **142***d* of the casing **140**, and the lower side is surrounded by the first member **148**.
(141) The partition wall **160** is a plate-like member here. The partition wall **160** is attached to, for example, a tube plate **132***a* disposed at the left end of the utilization heat exchanger **132**, but the attachment location is not limited. The tube plate **132***a* is a member for fixing a plurality of heat transfer tubes (not shown) of the utilization heat exchanger **132**. The partition wall **160** extends from the side wall **142***b* on the front side of the casing **140** to the partition plate **142***d* in the front-rear direction. The partition wall **160** extends vertically from the top panel of the casing **140** to the first member **148**. The partition wall **160** may come into contact with the side wall **142***b* on the front side of the casing **140**, the partition plate **142***d*, the top panel of the casing **140**, and the first

member **148** directly or indirectly via another member. Since the partition wall **160** and these members are in direct or indirect contact with each other, the flow of air between the first space S**1** and the second space S**2** is easily suppressed.
(142) An opening **144** through which the liquid refrigerant pipe **37***a* and the gas refrigerant pipe **37***b* pass is formed in the side wall **142***b* of the casing **140** that forms the first space S**1**, in other words, that surrounds the first space S**1**. The first space S**1** and the attic space CS in which the casing **140** is installed communicate with each other via the opening **144**. The first space S**1** and the attic space CS may communicate with each other via the opening **144**, but gaps between the liquid refrigerant pipe **37***a* and the gas refrigerant pipe **37***b* and the opening **144** may be closed by a sealing material or the like.
(143) The second expansion valve **34**, the first shutoff valve **52**, and the second shutoff valve **54** are disposed in the first space S**1** formed in this manner. The second expansion valve **34**, the first shutoff valve **52**, and the second shutoff valve **54** are disposed, for example, at a lower part of the first space S**1**. However, the present disclosure is not limited to this, and the position in which the second expansion valve **34**, the first shutoff valve **52**, and the second shutoff valve **54** are disposed in the first space S**1** may be appropriately determined. Further, the position of the first space S**1** described here is an example, and the first space S**1** may be formed at another place.
Second Embodiments
(1) Overall Outline
(144) An outline of an air conditioner **1100** according to one or more embodiments will be described with reference to FIGS. **8** and **9**. FIG. **8** is a schematic configuration diagram of the air conditioner **1100**. FIG. **9** is a control block diagram of the air conditioner **1100**. As described later, in one or more embodiments, the air conditioner **1100** includes a plurality of utilization units **1030** each having a second control unit **1038** and a plurality of shutoff valve devices **1060** each having a control unit **1062**. However, in FIG. **9**, only one second control unit **1038** and one control unit **1062** are illustrated in order to avoid complication of the drawing.
(145) The air conditioner **1100** performs a vapor compression refrigeration cycle to cool or heat the air conditioning target space **1000**R. The air conditioning target space **1000**R is, for example, a living room of an office or a house. In one or more embodiments, the air conditioner **1100** is capable of both cooling and heating the air conditioning target space **1000**R. However, the air conditioner of the present disclosure is not limited to an air conditioner capable of both cooling and heating, and may be, for example, a device capable of only cooling.
(146) The air conditioner **1100** mainly includes a heat source unit **1010** as an example of an air conditioning heat source unit, a utilization unit **1030** as an example of an air conditioning indoor unit, a gas-refrigerant connection pipe **1000**GP, a liquid-refrigerant connection pipe **1000**LP, and a shutoff valve device **1060**.
(147) In one or more embodiments, the air conditioner **1100** includes one heat source unit **1010**. However, the number of heat source units **1010** is not limited to one, and the air conditioner **1100** may include a plurality of heat source units **1010**.
(148) In one or more embodiments, the air conditioner **1100** includes three utilization units **1030**. However, the number of the utilization units **1030** is not limited to plural, and the air conditioner **1100** may include only one utilization unit **1030**. The air conditioner **1100** may include two or four or more utilization units **1030**.
(149) The gas-refrigerant connection pipe **1000**GP and the liquid-refrigerant connection pipe **1000**LP connect the heat source unit **1010** and the utilization unit **1030**. The gas-refrigerant connection pipe **1000**GP and the liquid-refrigerant connection pipe **1000**LP are laid at an installation site of the air conditioner **1100**. The pipe diameters and pipe lengths of the gas-refrigerant connection pipe **1000**GP and the liquid-refrigerant connection pipe **1000**LP are selected according to design specifications and installation environments. In the air conditioner **1100**, the heat source unit **1010** and the utilization unit **1030** are connected by the gas-refrigerant connection

pipe **1000**GP and the liquid-refrigerant connection pipe **1000**LP to form the refrigerant circuit **1000**RC. The refrigerant circuit **1000**RC includes a compressor **1012**, a heat-source heat exchanger **1016**, and a first expansion valve **1018** of the heat source unit **1010** to be described later, and a utilization heat exchanger **1032** and a second expansion valve **1034** of each utilization unit **1030** to be described later. The refrigerant circuit **1000**RC includes a first shutoff valve **1052** and a second shutoff valve **1054** of each shutoff valve device **1060** described later.
(150) The refrigerant circuit **1000**RC is filled with a refrigerant. Although not limited, the refrigerant sealed in the refrigerant circuit **1000**RC is flammable. Examples of the flammable refrigerant include refrigerants categorized in Class 3 (higher flammability), Class 2 (lower flammability), and Subclass 2L (slight flammability) in the standards according to ASHRAE 34 Designation and safety classification of refrigerant in the U.S.A. or the standards according to ISO 817 Refrigerants—Designation and safety classification. Examples of the refrigerant to be used herein may include, but not limited to, R1234yf, R1234ze(E), R516A, R445A, R444A, R454C, R444B, R454A, R455A, R457A, R459B, R452B, R454B, R447B, R32, R447A, R446A, and R459A. One or more embodiments adopt R32 as the refrigerant used therein. The air conditioner of the present disclosure is also useful when the refrigerant is not flammable.
(151) In one or more embodiments, the air conditioner **1100** includes three shutoff valve devices **1060**. Each shutoff valve device **1060** is provided corresponding to one of the utilization units **1030**.
(152) Each shutoff valve device **1060** includes a shutoff valve **1050** (i.e., shutoff valve group). The shutoff valve **1050** includes at least one of a first shutoff valve disposed in the liquid-refrigerant connection pipe **1000**LP and a second shutoff valve disposed in the gas-refrigerant connection pipe **1000**GP. In one or more embodiments, the shutoff valve **1050** of each shutoff valve device **1060** includes both the first shutoff valve **1052** disposed in the liquid-refrigerant connection pipe **1000**LP and the second shutoff valve **1054** disposed in the gas-refrigerant connection pipe **1000**GP.
(153) When closed, the first shutoff valve **1052** of each shutoff valve device **1060** shuts off the flow of the refrigerant flowing from the heat source unit **1010** or from the portion of the liquid-refrigerant connection pipe **1000**LP connecting the heat source unit **1010** and the first shutoff valve **1052** to the utilization unit **1030** corresponding to the shutoff valve device **1060** through the first shutoff valve **1052**.
(154) When closed, the second shutoff valve **1054** of each shutoff valve device **1060** shuts off the flow of the refrigerant flowing from the heat source unit **1010** or from the portion of the gas-refrigerant connection pipe **1000**GP connecting the heat source unit **1010** and the second shutoff valve **1054** to the utilization unit **1030** corresponding to the shutoff valve device **1060** through the second shutoff valve **1054**.
(2) Detailed Configuration
(155) The heat source unit **1010**, the utilization unit **1030**, and the shutoff valve device **1060** will be described in detail.
(156) (2-1) Heat Source Unit
(157) The heat source unit **1010** will be described with reference to FIGS. **8** and **9**.
(158) The heat source unit **1010** is installed, for example, on a rooftop of a building in which the air conditioner **1100** is installed, in a machine room of the building, or around the building or the like. In the heat source unit **1010**, heat is exchanged between the heat source and the refrigerant in a heat-source heat exchanger **1016** described later. In one or more embodiments, air is used as the heat source, but the present disclosure is not limited thereto, and liquid such as water may be used as the heat source.
(159) The heat source unit **1010** mainly includes a compressor **1012**, a flow direction switching mechanism **1014**, a heat-source heat exchanger **1016**, a first expansion valve **1018**, a first fan **1020**, a first stop valve **1024**, a second stop valve **1026**, and a first control unit **1022** (see FIGS. **8** and **9**). The configuration of the heat source unit **1010** is merely an example. The heat source unit **1010**

may not have a part of the illustrated configuration or may have a configuration other than the illustrated configuration as long as the air conditioner **1100** can function.
(160) The heat source unit **1010** includes, as refrigerant pipes, a suction pipe **1011***a*, a discharge pipe **1011***b*, a first gas refrigerant pipe **1011***c*, a liquid refrigerant pipe **1011***d*, and a second gas refrigerant pipe **1011***e* (see FIG. **8**). The suction pipe **1011***a* connects the flow direction switching mechanism **1014** and the suction side of the compressor **1012**. The discharge pipe **1011***b* connects the discharge side of the compressor **1012** and the flow direction switching mechanism **1014**. The first gas refrigerant pipe **1011***c* connects the flow direction switching mechanism **1014** and the gas side end of the heat-source heat exchanger **1016**. The liquid refrigerant pipe **1011***d* connects the liquid side end of the heat-source heat exchanger **1016** and the liquid-refrigerant connection pipe **1000**LP. A first stop valve **1024** is provided at a connection portion between the liquid refrigerant pipe **1011***d* and the liquid-refrigerant connection pipe **1000**LP. The first expansion valve **1018** is provided between the heat-source heat exchanger **1016** and the first stop valve **1024** of the liquid refrigerant pipe **1011***d*. The second gas refrigerant pipe **1011***e* connects the flow direction switching mechanism **1014** and the gas-refrigerant connection pipe **1000**GP. A second stop valve **1026** is provided at a connection portion between the second gas refrigerant pipe **1011***e* and the gas-refrigerant connection pipe **1000**GP.
(161) (2-1-1) Compressor
(162) The compressor **1012** sucks and compresses a low-pressure gas refrigerant in the refrigeration cycle and discharges a high-pressure gas refrigerant in the refrigeration cycle. The compressor **1012** is, for example, an inverter control-type compressor. However, the compressor **1012** may be a constant-speed compressor.
(163) (2-1-2) Flow Direction Switching Mechanism
(164) The flow direction switching mechanism **1014** is a mechanism that switches the flow direction of the refrigerant in the refrigerant circuit **1000**RC according to the operation mode (cooling operation mode/heating operation mode) of the air conditioner **1100**. The flow direction switching mechanism **1014** is a four-way switching valve.
(165) In the cooling operation mode, the flow direction switching mechanism **1014** switches the flow direction of the refrigerant in the refrigerant circuit **1000**RC such that the refrigerant discharged from the compressor **1012** is sent to the heat-source heat exchanger **1016**. Specifically, in the cooling operation mode, the flow direction switching mechanism **1014** causes the suction pipe **1011***a* to communicate with the second gas refrigerant pipe **1011***e* and causes the discharge pipe **1011***b* to communicate with the first gas refrigerant pipe **1011***c* (see the solid line in FIG. **8**). In the cooling operation mode, the heat-source heat exchanger **1016** functions as a condenser, and the utilization heat exchanger **1032** functions as an evaporator.
(166) In the heating operation mode, the flow direction switching mechanism **1014** switches the flow direction of the refrigerant in the refrigerant circuit **1000**RC such that the refrigerant discharged from the compressor **1012** is sent to the use heat exchanger **1032**. Specifically, in the heating operation mode, the flow direction switching mechanism **1014** causes the suction pipe **1011***a* to communicate with the first gas refrigerant pipe **1011***c* and causes the discharge pipe **1011***b* to communicate with the second gas refrigerant pipe **1011***e* (see a broken line in FIG. **8**). In the heating operation mode, the heat-source heat exchanger **1016** functions as an evaporator, and the utilization heat exchanger **1032** functions as a condenser.
(167) The flow direction switching mechanism **1014** may be realized without using a four-way switching valve. For example, the flow direction switching mechanism **1014** may be configured by combining a plurality of electromagnetic valves and pipes so as to realize switching of the refrigerant flow direction as described above.
(168) (2-1-3) Heat-Source Heat Exchanger
(169) In the heat-source heat exchanger **1016**, heat is exchanged between the refrigerant flowing through the heat-source heat exchanger **1016** and air as a heat source. The heat-source heat

exchanger **1016** functions as a condenser (radiator) of the refrigerant during the cooling operation, and functions as an evaporator of the refrigerant during the heating operation. Although not limited, the heat-source heat exchanger **1016** is, for example, a fin-and-tube heat exchanger having a plurality of heat transfer tubes and a plurality of heat transfer fins.
(170) (2-1-4) First Expansion Valve
(171) The first expansion valve **1018** is a mechanism that decompresses the refrigerant and adjusts the flow rate of the refrigerant. In one or more embodiments, the first expansion valve **1018** is an electronic expansion valve whose opening degree is adjustable. The opening degree of the first expansion valve **1018** is appropriately adjusted according to the operation situation. The first expansion valve **1018** is not limited to the electronic expansion valve, and may be another type of valve such as an automatic temperature expansion valve.
(172) (2-1-5) First Fan
(173) The first fan **1020** is a blower that generates an air flow that flows into the heat source unit **1010** from the outside of the heat source unit **1010**, passes through the heat-source heat exchanger **1016**, and then flows out of the heat source unit **1010**. The first fan **1020** is, for example, an inverter control-type fan. However, the first fan **1020** may be a constant-speed fan.
(174) (2-1-6) First Stop Valve and Second Stop Valve
(175) The first stop valve **1024** is a valve provided at a connection portion between the liquid refrigerant pipe **1011***d* and the liquid-refrigerant connection pipe **1000**LP. The second stop valve **1026** is a valve provided at a connection portion between the second gas refrigerant pipe **1011***e* and the gas-refrigerant connection pipe **1000**GP. The first stop valve **1024** and the second stop valve **1026** are manual valves. The first stop valve **1024** and the second stop valve **1026** are opened when the air conditioner **1100** is used.
(176) (2-1-7) First Control Unit
(177) The first control unit **1022** controls operations of various devices of the heat source unit **1010**. The first control unit **1022** mainly includes a microcontroller unit (MCU), various electric circuits, and electronic circuits (not shown). The MCU includes a CPU, a memory, an I/O interface, and the like. The memory of the MCU stores various programs to be executed by the CPU of the MCU. Note that the various functions of the first control unit **1022** need not be implemented by software, and may be implemented by hardware or may be implemented by cooperation of hardware and software.
(178) The first control unit **1022** is electrically connected to various devices of the heat source unit **1010** including the compressor **1012**, the flow direction switching mechanism **1014**, the first expansion valve **1018**, and the first fan **1020** (see FIG. **9**). The first control unit **1022** is electrically connected to various sensors (not shown) provided in the heat source unit **1010**. Although not limited, the sensor provided in the heat source unit **1010** includes a temperature sensor and a pressure sensor provided in the discharge pipe **1011***b* and the suction pipe **1011***a*, a temperature sensor provided in the heat-source heat exchanger **1016** and the liquid refrigerant pipe **1011***d*, a temperature sensor that measures the temperature of the heat source air, and the like. The heat source unit **1010** may include all or some of these sensors.
(179) As illustrated in FIG. **9**, the first control unit **1022** is connected to the second control unit **1038** of the utilization unit **1030** via a communication line. The first control unit **1022** and the second control unit **1038** exchange various signals via a communication line. The first control unit **1022** and the second control unit **1038** cooperate to function as a controller **1090** that controls the operation of the air conditioner **1100**. The function of the controller **1090** will be described later.
(180) (2-2) Utilization Unit
(181) The utilization unit **1030** will be described. In the utilization unit **1030**, heat is exchanged between the refrigerant and the air in the air conditioning target space **1000**R in the utilization heat exchanger **1032** described later, and as a result, cooling or heating of the air conditioning target space **1000**R is performed.

(182) In one or more embodiments, the air conditioner **1100** includes three utilization units **1030**. Structures and capabilities of the three utilization units **1030** may be the same or different from each other. Here, the utilization units **1030** will be described as having the same configuration.
(183) The type of the utilization unit **1030** is, for example, a wall-mounted type as illustrated in FIG. **10**A, and is attached to the wall of the air conditioning target space **1000**R. The type of the utilization unit **1030** may be, for example, a floor type as illustrated in FIG. **10**B, and may be installed on the floor of the air conditioning target space **1000**R. The type of the utilization unit **1030** may be, for example, a ceiling suspended type as illustrated in FIG. **10**C, and may be installed to be suspended on the ceiling of the air conditioning target space **1000**R. The air conditioner **1100** may include two or more types of utilization units **1030**.
(184) As shown in FIGS. **8**, **9**, and **10**A to **10**C, the utilization unit **1030** mainly includes a casing **1042**, a utilization heat exchanger **1032**, a second expansion valve **1034**, a second fan **1036**, a refrigerant detector **1040**, and a second control unit **1038**. Note that the configuration of the utilization unit **1030** illustrated here is merely an example. The utilization unit **1030** may not have a part of the illustrated configuration or may have a configuration other than the illustrated configuration as long as the air conditioner **1100** can function.
(185) The utilization unit **1030** includes, as refrigerant pipes, a liquid refrigerant pipe **1037***a* and a gas refrigerant pipe **1037***b* connected to the utilization heat exchanger **1032** (see FIG. **8**). The liquid refrigerant pipe **1037***a* connects the liquid-refrigerant connection pipe **1000**LP and the liquid side of the utilization heat exchanger **1032**. The gas refrigerant pipe **1037***b* connects the gas-refrigerant connection pipe **1000**GP and the gas side of the utilization heat exchanger **1032**. The liquid refrigerant pipe **1037***a* is provided with a second expansion valve **1034**.
(186) (2-2-1) Casing
(187) The casing **1042** accommodates therein various devices of the utilization unit **1030** including the utilization heat exchanger **1032**, the second expansion valve **1034**, and the second fan **1036**. The casing **1042** is disposed in the air conditioning target space **1000**R as shown in FIGS. **10**A to **10**C. Unlike a utilization unit of a ceiling-embedded type or the like, a part of the casing **1042** is not arranged in the attic space **1000**S.
(188) The casing **1042** is provided with a suction port (not shown) for taking in the air in the air conditioning target space **1000**R. The casing **1042** has a blow-out port (not shown) through which air introduced into the casing **1042** through the suction port and having exchanged heat with the refrigerant in the utilization heat exchanger **1032** is blown out into the air conditioning target space **1000**R. The shape and structure of the casing **1042** vary depending on the type (wall-mounted type, floor type, ceiling-suspended type) of the utilization unit **1030**. Here, description of the shape and structure of the casing **1042** of each type of utilization unit **1030** is omitted.
(189) (2-2-2) Indoor Heat Exchanger
(190) In the utilization heat exchanger **1032**, heat is exchanged between the refrigerant flowing through the utilization heat exchanger **1032** and air. The utilization heat exchanger **1032** functions as an evaporator of the refrigerant during the cooling operation, and functions as a condenser (radiator) of the refrigerant during the heating operation. Although not limited, the utilization heat exchanger **1032** is, for example, a fin-and-tube heat exchanger having a plurality of heat transfer tubes and a plurality of heat transfer fins.
(191) (2-2-3) Second Expansion Valve
(192) The second expansion valve **1034** is a mechanism that decompresses the refrigerant and adjusts the flow rate of the refrigerant. In one or more embodiments, the second expansion valve **1034** is an electronic expansion valve whose opening degree is adjustable. The opening degree of the second expansion valve **1034** is appropriately adjusted according to the operation situation. The second expansion valve **1034** is not limited to the electronic expansion valve, and may be another type of valve such as an automatic temperature expansion valve.
(193) (2-2-4) Second Fan

(194) The second fan **1036** is a blower that generates an air flow that flows into the casing **1042** from a suction port (not shown) of the casing **1042**, passes through the utilization heat exchanger **1032**, and then flows out of the casing **1042** from a blow-out port of the casing **1042**. The second fan **1036** is, for example, an inverter control-type fan. However, the second fan **1036** may be a constant-speed fan.
(195) (2-2-5) Refrigerant Detector
(196) The refrigerant detector **1040** is a sensor that detects the leakage of refrigerant at the utilization unit **1030**. The refrigerant detector **1040** is provided, for example, in the casing **1042** of the utilization unit **1030**. The refrigerant detector **1040** may be disposed outside the casing **1042** of the utilization unit **1030**. A plurality of refrigerant detectors **1040** may be provided.
(197) The refrigerant detector **1040** is, for example, a semiconductor-type sensor. The semiconductor refrigerant detector **1040** includes a semiconductor-type detection element (not shown). The semiconductor detector element has electric conductivity that changes depending on whether it is in a case where no refrigerant gas exists therearound and in a case where refrigerant gas exists therearound. When the refrigerant gas exists around the semiconductor-type detection element, the refrigerant detector **1040** outputs a relatively large current as a detection signal. On the other hand, when the refrigerant gas does not exist around the semiconductor-type detection element, the refrigerant detector **1040** outputs a relatively small current as a detection signal.
(198) The type of the refrigerant detector **1040** is not limited to the semiconductor type, and may be any sensor capable of detecting refrigerant gas. For example, the refrigerant detector **1040** may be an infrared sensor that outputs a detection signal according to a detection result of the refrigerant.
(199) (2-2-6) Second Control Unit
(200) The second control unit **1038** controls operations of various devices of the utilization unit **1030**. The second control unit **1038** includes a microcontroller unit (MCU), various electric circuits, and electronic circuits (not shown). The MCU includes a CPU, a memory, an I/O interface, and the like. The memory of the MCU stores various programs to be executed by the CPU of the MCU. Note that the various functions of the second control unit **1038** does not need to be implemented by software, and may be implemented by hardware or may be implemented by cooperation of hardware and software.
(201) The second control unit **1038** is electrically connected to various devices of the utilization unit **1030** including the second expansion valve **1034** and the second fan **1036** (see FIG. **9**). The second control unit **1038** is electrically connected to the refrigerant detector **1040**. The second control unit **1038** is electrically connected to a sensor (not shown) provided in the utilization unit **1030**. Although not limited, the sensors (not shown) include a temperature sensor provided in the utilization heat exchanger **1032** and the liquid refrigerant pipe **1037***a*, a temperature sensor that measures the temperature of the air conditioning target space **1000**R, and the like. The utilization unit **1030** may include all or some of these sensors.
(202) The second control unit **1038** is communicably connected to the control unit **1062** that controls the operations of the first shutoff valve **1052** and the second shutoff valve **1054** of the shutoff valve device **1060** by a communication line (see FIG. **9**).
(203) As illustrated in FIG. **9**, the second control unit **1038** is connected to the first control unit **1022** of the heat source unit **1010** via a communication line. The second control unit **1038** is communicably connected to a remote control unit for operating the air conditioner **1100** (not shown) via a communication line. The first control unit **1022** and the second control unit **1038** cooperate to function as a controller **1090** that controls the operation of the air conditioner **1100**.
(204) The function of the controller **1090** will be described. Some or all of various functions of the controller **1090** described below may be executed by a control device provided separately from the first control unit **1022** and the second control unit **1038**.
(205) The controller **1090** controls the operation of the flow direction switching mechanism **1014** such that the heat-source heat exchanger **1016** functions as a condenser of the refrigerant and the

utilization heat exchanger **1032** functions as an evaporator of the refrigerant during the cooling operation. During the heating operation, the controller **1090** controls the operation of the flow direction switching mechanism **1014** such that the heat-source heat exchanger **1016** functions as an evaporator of the refrigerant and the utilization heat exchanger **1032** functions as a condenser of the refrigerant. The controller **1090** operates the compressor **1012**, the first fan **1020**, and the second fan **1036** during the cooling operation and the heating operation. During the cooling operation and the heating operation, the controller **1090** adjusts the numbers of rotations of the motors of the compressor **1012**, the first fan **1020**, and the second fan **1036** and the opening degrees of the first expansion valve **1018** and the second expansion valve **1034** based on various instructions (set temperature, set air volume, and the like) input to the remote control unit and measurement values of various temperature sensors and pressure sensors. Various control modes are generally known for the control of the operations of the various devices of the air conditioner **1100** during the cooling operation and the heating operation, and thus, the description thereof will be omitted here.
(206) The control of the air conditioner **1100** by the controller **1090** when the refrigerant is detected by the refrigerant detector **1040** of any of the utilization units **1030** will be described later.
(207) (2-3) Shutoff Valve Device
(208) Each shutoff valve device **1060** is installed corresponding to one of the utilization units **1030**. The shutoff valve device **1060** is a device that closes the shutoff valve **1050** included in the shutoff valve device **1060** to suppress the inflow of the refrigerant into the utilization unit **1030** corresponding to the shutoff valve device **1060**.
(209) The shutoff valve device **1060** mainly includes a shutoff valve **1050**, a main body casing **1064**, an electric component **1062***a*, and an electric component box **1066** that accommodates the electric component **1062***a*. The electric component **1062***a* includes a control unit **1062** that controls operations of the first shutoff valve **1052** and the second shutoff valve **1054**.
(210) (2-3-1) Shutoff Valve
(211) The shutoff valve **1050** includes at least one of a first shutoff valve disposed in the liquid-refrigerant connection pipe **1000**LP and a second shutoff valve disposed in the gas-refrigerant connection pipe **1000**GP. In one or more embodiments, the shutoff valve **1050** of each shutoff valve device **1060** includes both the first shutoff valve **1052** disposed in the liquid-refrigerant connection pipe **1000**LP and the second shutoff valve **1054** disposed in the gas-refrigerant connection pipe **1000**GP.
(212) A liquid-refrigerant connection pipe **1000**LP connecting the heat source unit **1010** and the first shutoff valve **1052** is connected to one end of the first shutoff valve **1052** of the shutoff valve device **1060**. A liquid-refrigerant connection pipe **1000**LP that connects the liquid refrigerant pipe **1037***a* of the utilization unit **1030** corresponding to the shutoff valve device **1060** and the first shutoff valve **1052** is connected to the other end of the first shutoff valve **1052** of the shutoff valve device **1060**.
(213) A gas-refrigerant connection pipe **1000**GP connecting the heat source unit **1010** and the second shutoff valve **1054** is connected to one end of the second shutoff valve **1054** of the shutoff valve device **1060**. A gas-refrigerant connection pipe **1000**GP that connects the gas refrigerant pipe **1037***b* of the utilization unit **1030** corresponding to the shutoff valve device **1060** and the second shutoff valve **1054** is connected to the other end of the second shutoff valve **1054** of the shutoff valve device **1060**.
(214) The first shutoff valve **1052** and the second shutoff valve **1054** are valves that suppress the leakage of refrigerant into the air conditioning target space **1000**R when the leakage of refrigerant occurs at the utilization unit **1030**. The first shutoff valve **1052** and the second shutoff valve **1054** are, for example, electromagnetic valves capable of switching between a closed state (fully closed) and an open state (fully open). However, the types of the first shutoff valve **1052** and the second shutoff valve **1054** are not limited to the electromagnetic valve, and may be, for example, an electric valve.

(215) The first shutoff valve **1052** and the second shutoff valve **1054** of the shutoff valve device **1060** are normally opened. The normal time here means a time when the second control unit **1038** of the utilization unit **1030** corresponding to the shutoff valve device **1060** has not transmitted a signal instructing the control unit **1062** to close the shutoff valve **1050**.
(216) On the other hand, when the refrigerant detector **1040** of the utilization unit **1030** corresponding to the shutoff valve device **1060** detects the refrigerant, the first shutoff valve **1052** and the second shutoff valve **1054** are closed. Specifically, when the refrigerant detector **1040** of the utilization unit **1030** corresponding to the shutoff valve device **1060** detects the refrigerant and the second control unit **1038** of the corresponding utilization unit **1030** transmits a signal instructing to close the shutoff valve **1050** to the control unit **1062** of the corresponding shutoff valve device **1060**, the control unit **1062** that has received the signal performs control to close the first shutoff valve **1052** and the second shutoff valve **1054**. When the first shutoff valve **1052** and the second shutoff valve **1054** of the shutoff valve device **1060** are closed, the inflow of the refrigerant from the heat source unit **1010**, the pipe connecting the heat source unit **1010** and the first shutoff valve **1052**, and the pipe connecting the heat source unit **1010** and the second shutoff valve **1054** to the utilization unit **1030** corresponding to the shutoff valve device **1060** is suppressed.
(217) As illustrated in FIGS. **10**A to **10**C, the shutoff valve **1050** (the first shutoff valve **1052** and the second shutoff valve **1054**) is disposed in the attic space **1000**S above the ceiling **1000**CL of the air conditioning target space **1000**R. The shutoff valve **1050** may be disposed in the attic space **1000**S and in the vicinity of the corresponding utilization unit **1030**. Although the arrangement is not limited, the shutoff valve **1050** is arranged, for example, in the attic space **1000**S and in the vicinity immediately above the corresponding utilization unit **1030**.
(218) When the upper floor is present above the air conditioning target space **1000**R in the building in which the air conditioner **1100** is installed, the attic space **1000**S is a space between the ceiling **1000**CL of the air conditioning target space **1000**R and the floor of the upper floor (one floor above) of the air conditioning target space **1000**R. When the air conditioning target space **1000**R is the uppermost floor in the building in which the air conditioner **1100** is installed (when there is no upper floor above the air conditioning target space **1000**R), the attic space **1000**S is a space between the ceiling **1000**CL of the air conditioning target space **1000**R and the roof of the building.
(219) The attic space **1000**S is a space partitioned from the air conditioning target space **1000**R by the building material constituting the ceiling **1000**CL. The fact that the attic space **1000**S and the air conditioning target space **1000**R are partitioned does not mean that both spaces are partitioned in an airtight state, but means that the flow of air between both spaces is at least suppressed by the building material constituting the ceiling **1000**CL. For example, some air may flow between the attic space **1000**S and the air conditioning target space **1000**R via a gap between building materials. The gap between the building materials is not limited, but is, for example, a gap between an inspection port provided in the ceiling **1000**CL for inspecting the attic space **1000**S and a closing member that closes the inspection port.
(220) In the air conditioner **1100** of one or more embodiments, since the shutoff valve **1050** is installed in the attic space **1000**S, even if the refrigerant leaks around the shutoff valve **1050**, the refrigerant flows not into the air conditioning target space **1000**R but into the attic space **1000**S partitioned from the air conditioning target space **1000**R. In short, in the air conditioner **1100** according to one or more embodiments, the refrigerant is less likely to flow into the air conditioning target space **1000**R where people are active. Therefore, the air conditioner **1100** has high safety even when, for example, a flammable refrigerant is used.
(221) (2-3-2) Main Body Casing
(222) The main body casing **1064** is a casing that accommodates the shutoff valve **1050**. Specifically, the main body casing **1064** is a casing that accommodates the first shutoff valve **1052** and the second shutoff valve **1054**.

(223) As illustrated in FIG. **11**A, the main body casing **1064** is installed in the attic space **1000**S. In order to dispose the shutoff valve **1050** in the vicinity of the corresponding utilization unit **1030**, the main body casing **1064** may be installed in the vicinity of the corresponding utilization unit **1030**. Although not limited, the main body casing **1064** is installed in the attic space **1000**S and in the vicinity immediately above the corresponding utilization unit **1030**.
(224) For example, as illustrated in FIG. **11**A, the main body casing **1064** is formed with an opening **1064***a* through which a liquid-refrigerant connection pipe **1000**LP connected to both ends of the first shutoff valve **1052** and a gas-refrigerant connection pipe **1000**GP connected to both ends of the second shutoff valve **1054** are inserted. In FIG. **11**A, a part of these openings **1064***a* (for example, an opening **1064***a* through which the liquid-refrigerant connection pipe **1000**LP connecting the heat source unit **1010** and the first shutoff valve **1052** and the gas-refrigerant connection pipe **1000**GP connecting the heat source unit **1010** and the second shutoff valve **1054** pass) is illustrated. In the example of FIG. **11**A, a plurality of refrigerant pipes (one liquid-refrigerant connection pipe **1000**LP and one gas-refrigerant connection pipe **1000**GP) are arranged to extend through one opening **1064***a*.
(225) In place of the openings through which the plurality of refrigerant pipes pass, the main body casing **1064** may include, as shown in FIG. **11**B, openings **1064***a* arrange so that one refrigerant pipe (one liquid-refrigerant connection pipe **1000**LP or one gas-refrigerant connection pipe **1000**GP) extends through each of the opening **1064***a*.
(226) The opening **1064***a* may be provided with a heat insulating material **1068** that closes a gap between the opening **1064***a* and the liquid-refrigerant connection pipe **1000**LP, a gap between the opening **1064***a* and the gas-refrigerant connection pipe **1000**GP, and a gap between the liquid-refrigerant connection pipe **1000**LP and the gas-refrigerant connection pipe **1000**GP. By closing the gap between the opening **1064***a* and the connection pipes **1000**LP and **1000**GP and the gap between the refrigerant pipes with the heat insulating material **1068** in this manner, the leakage of the refrigerant into the attic space **1000**S is suppressed even if the refrigerant leaks inside the main body casing **1064**, and safety is therefore high.
(227) (2-3-3) Electric Component
(228) The electric component **1062***a* includes various components for operating the first shutoff valve **1052** and the second shutoff valve **1054**. Although not limited, the electric component **1062***a* includes, for example, a switching unit capable of switching the flow of current, such as a printed circuit board, an electromagnetic relay, and a switching element, a terminal block to which power is supplied, and an input unit to which a signal from the second control unit **1038** is input. The electric component **1062***a* is electrically connected to the first shutoff valve **1052** and the second shutoff valve **1054** by an electric wire for supplying a drive voltage. At least a part of the electric component **1062***a* functions as the control unit **1062** that closes the first shutoff valve **1052** and the second shutoff valve **1054** in response to a signal requesting closing of the shutoff valve **1050** from the second control unit **1038** of the utilization unit **1030**.
(229) The control unit **1062** includes, for example, a microcontroller unit (MCU), various electric circuits, and electronic circuits (not shown). The MCU includes a CPU, a memory, an I/O interface, and the like. The memory of the MCU stores various programs to be executed by the CPU of the MCU. Note that the various functions of the control unit **1062** does not need to be implemented by software, and may be implemented by hardware or may be implemented by cooperation of hardware and software.
(230) The electric component **1062***a* is accommodated in the electric component box **1066**. The electric component box **1066** may be disposed outside the main body casing **1064**. The electric component box **1066** is installed, for example, in the attic space **1000**S. When the electric component box **1066** and the main body casing **1064** have independent configurations, the electric component box **1066** may not be disposed in the vicinity of the main body casing **1064**. The installation position of the electric component box **1066** may be appropriately determined.

(3) Control of Air Conditioner by Controller at Time of Refrigerant Detection

(231) The control of the air conditioner **1100** by the controller **1090** when the refrigerant is detected by the refrigerant detector **1040** of any of the utilization units **1030** will be described. Here, the case where the refrigerant is detected by the refrigerant detector **1040** means a case where the value of the current output as the detection signal by the refrigerant detector **1040** is larger than a predetermined threshold.

(232) When the refrigerant is detected by the refrigerant detector **1040** of any of the utilization units **1030**, the controller **1090** transmits a signal instructing to close the shutoff valve **1050** of the shutoff valve device **1060** to the control unit **1062** of the shutoff valve device **1060** corresponding to the utilization unit **1030** in which the refrigerant is detected. The signal instructing to close the shutoff valve **1050** may be a contact signal. The control unit **1062** of the shutoff valve device **1060** closes the shutoff valve **1050** (in one or more embodiments, the first shutoff valve **1052** and the second shutoff valve **1054**) based on this signal.

(233) When the refrigerant is detected by the refrigerant detector **1040** of any of the utilization units **1030**, the controller **1090** may notify of the leakage of refrigerant using an alarm (not shown) in addition to transmitting a signal instructing to close the shutoff valve **1050** to the control unit **1062** of the shutoff valve device **1060**.

(234) When the refrigerant is detected by the refrigerant detector **1040** of any of the utilization units **1030**, the controller **1090** may stop the operation of the entire air conditioner **1100** by stopping the operation of the compressor **1012** and in addition to transmitting a signal instructing to close the shutoff valve **1050** to the control unit **1062** of the shutoff valve device **1060**.

(235) When the refrigerant is detected by the refrigerant detector **1040** of any of the utilization units **1030**, the controller **1090** may transmit a signal instructing to close the shutoff valve **1050** to the control unit **1062** of the shutoff valve device **1060** corresponding to that utilization unit **1030** and may further transmit a signal instructing to close the shutoff valve **1050** to the control unit **1062** of another shutoff valve device **1060** (for example, all the shutoff valve devices **1060**).

(4) Features

(236) (4-1)

(237) The air conditioner **1100** of one or more embodiments includes a utilization unit **1030** as an air conditioning indoor unit, a heat source unit **1010** as an air conditioning heat source unit, and a shutoff valve device **1060**. The utilization unit **1030** is installed in the air conditioning target space **1000**R. The heat source unit **1010** is connected to the utilization unit **1030** via a liquid-refrigerant connection pipe **1000**LP and a gas-refrigerant connection pipe **1000**GP. The shutoff valve device **1060** includes a shutoff valve **1050** disposed in the attic space **1000**S above the ceiling **1000**CL of the air conditioning target space **1000**R. The shutoff valve **1050** includes at least one of a first shutoff valve **1052** disposed in the liquid-refrigerant connection pipe **1000**LP and a second shutoff valve **1054** disposed in the gas-refrigerant connection pipe **1000**GP. In one or more embodiments, the shutoff valve **1050** includes both the first shutoff valve **1052** and the second shutoff valve **1054**.

(238) As disclosed in Patent Literature 2 (JP 2013-19621 A), there is known an air conditioner in which an air conditioning indoor unit is provided with a shutoff valve for preventing the leakage of refrigerant separately from the air conditioning indoor unit. The shutoff valve is disposed in the vicinity of the air conditioning indoor unit to reduce the amount of refrigerant leaking from the air conditioning indoor unit when the leakage of refrigerant is detected.

(239) However, when the shutoff valve is disposed in the air conditioning target space adjacent to the air conditioning indoor unit disposed in the air conditioning target space, if the refrigerant leaks from the shutoff valve, a relatively large amount of refrigerant may leak into the air conditioning target space.

(240) In the air conditioner **1100**, even if the refrigerant leaks around the shutoff valve **1050**, the refrigerant flows not into the air conditioning target space **1000**R but into the attic space **1000**S partitioned from the air conditioning target space **1000**R, and thus safety is high.

(241) (4-2)
(242) In the air conditioner **1100** of one or more embodiments, the shutoff valve device **1060** includes a main body casing **1064** as an example of a casing that accommodates the first shutoff valve **1052** and the second shutoff valve **1054**.
(243) In the present air conditioner **1100**, the shutoff valve device **1060** is a unit in which the first shutoff valve **1052**, the second shutoff valve **1054**, and the main body casing **1064** accommodating them are unitized. It is therefore easy to incorporate the shutoff valve device **1060** into the air conditioner **1100**.
(244) (4-3)
(245) In the air conditioner **1100** of one or more embodiments, the shutoff valve device **1060** includes an electric component box **1066** that accommodates electric components **1062***a* for operating the shutoff valve **1050**. The electric component box **1066** is disposed outside the main body casing **1064**.
(246) In the present air conditioner **1100**, since the electric component box **1066** is disposed outside the main body casing **1064**, if the refrigerant is flammable, and the refrigerant leaks around the shutoff valve **1050**, it is possible to suppress contact between the refrigerant and the electric component **1062***a* that can be an ignition source.
(247) (4-4)
(248) In the air conditioner **1100** of one or more embodiments, an opening **1064***a* is formed in the main body casing **1064**. The liquid-refrigerant connection pipe **1000**LP connected to the first shutoff valve **1052** and the gas-refrigerant connection pipe **1000**GP connected to the second shutoff valve **1054** extend through the opening **1064***a* of the main body casing **1064**. The shutoff valve device **1060** includes a heat insulating material **1068** that closes a gap between the opening **1064***a* and the liquid-refrigerant connection pipe **1000**LP and a gap between the opening **1064***a* and the gas-refrigerant connection pipe **1000**GP.
(249) In the present air conditioner **1100**, since the gap between the opening **1064***a* and the refrigerant connection pipes **1000**LP and **1000**GP extending through the opening **1064***a* is closed by the heat insulating material **1068**, the leakage of refrigerant into the attic space **1000**S is suppressed even if the refrigerant leaks inside the main body casing **1064**, and safety is high.
(5) Modifications
(250) The above described embodiments may be appropriately modified as described in the following modifications. Some or all of the modifications may be combined with the above-described embodiments or another modification as long as no contradiction occurs.
(251) (5-1) Modification 2A
(252) In the above-described embodiments, the control unit **1062** of the shutoff valve device **1060** controls the operation of the shutoff valve **1050**, but the present disclosure is not limited to such a configuration. For example, the shutoff valve device **1060** may not include the control unit **1062**, and the controller **1090** of the air conditioner **1100**, more specifically, the second control unit **1038** of the utilization unit **1030**, for example, may control the operation of the shutoff valve **1050**.
(253) (5-2) Modification 2B
(254) In the above-described embodiments, the shutoff valve device **1060** includes, as the shutoff valve **1050**, the first shutoff valve **1052** and the second shutoff valve **1054** dedicated for preventing refrigerant leakage. However, in the shutoff valve device **1060**, a valve used for a purpose other than the purpose of preventing the refrigerant leakage may be used as the shutoff valve **1050**.
(255) For example, the shutoff valve device **1060***a* of the air conditioner **1100** illustrated in FIG. **12** does not include the first shutoff valve **1052**. The utilization unit **1030***a* of the air conditioner **1100** illustrated in FIG. **12** does not include the second expansion valve **1034**, and instead, the shutoff valve device **1060***a* includes the second expansion valve **1034** as the shutoff valve **1050**. In short, the shutoff valve device **1060***a* includes the second expansion valve **1034** and the second shutoff valve **1054** as the shutoff valve **1050**.

(256) In the air conditioner **1100** illustrated in FIG. **12**, the controller **1090** of the air conditioner **1100** also functions as a control unit of the shutoff valve device **1060***a*. However, the present disclosure is not limited to such a configuration, and the shutoff valve device **1060***a* may include a control unit that controls operations of the second expansion valve **1034** and the second shutoff valve **1054**. During the cooling operation and the heating operation, the controller **1090** adjusts the opening degree of the second expansion valve **1034** based on various instructions (set temperature, set air volume, and the like) input to the remote control unit and measurement values of various temperature sensors and pressure sensors. When the refrigerant is detected by the refrigerant detector **1040** of any of the utilization units **1030**, the controller **1090** closes the second expansion valve **1034** and the second shutoff valve **1054** of the shutoff valve device **1060** corresponding to the utilization unit **1030** in which the refrigerant is detected.
(257) (5-3) Modification 2C
(258) In the above-described embodiments, the shutoff valve device **1060** includes, as the shutoff valve **1050**, both the first shutoff valve **1052** disposed in the liquid-refrigerant connection pipe **1000**LP and the second shutoff valve **1054** disposed in the gas-refrigerant connection pipe **1000**GP. However, as illustrated in FIG. **13**, the shutoff valve device **1060***b* of the air conditioner **1100** may include only the second shutoff valve **1054** as the shutoff valve **1050**.
(259) Here, when the refrigerant is detected by the refrigerant detector **1040** of any of the utilization units **1030**, the controller **1090** transmits an instruction signal to the control unit **1062** of the shutoff valve device **1060***b* corresponding to the utilization unit **1030** in which the refrigerant is detected so as to close the second shutoff valve **1054**. Further, when the refrigerant detector **1040** of any of the utilization units **1030** detects the refrigerant, the controller **1090** may close the second expansion valve **1034** of the utilization unit **1030** in which the refrigerant is detected.
(260) (5-4) Modification 2D
(261) In the above-described embodiments, the shutoff valve device **1060** includes the main body casing **1064** that accommodates the shutoff valve **1050** therein, but the present disclosure is not limited thereto. The shutoff valve device **1060** may not include the main body casing **1064**, and the shutoff valve **1050** may be disposed in the attic space **1000**S as it is.
(262) The electric component **1062***a* may also be disposed in the attic space **1000**S as it is, instead of in the electric component box **1066**.
(263) (5-5) Modification 2E
(264) In the above-described embodiments, one shutoff valve device **1060** is provided for each utilization unit **1030**, but the present disclosure is not limited thereto. For example, the shutoff valve device **1060** may be a device in which the shutoff valves **1050** for the plurality of utilization units **1030** are accommodated in one main body casing **1064**.
(265) (5-6) Modification 2F
(266) In the above-described embodiments, one first shutoff valve **1052** and one second shutoff valve **1054** are provided for each utilization unit **1030**, but the present disclosure is not limited thereto.
(267) For example, in the air conditioner, one first shutoff valve and one second shutoff valve may be provided in each of the liquid refrigerant pipe and the gas refrigerant pipe before being branched to supply the refrigerant to a plurality of utilization units **1030** (referred to as a utilization unit group). When the refrigerant is detected by one refrigerant detector **1040** of the utilization units belonging to the utilization unit group, the inflow of the refrigerant into the plurality of utilization units **1030** belonging to the utilization unit group may be suppressed by closing the first shutoff valve and the second shutoff valve. In other words, the shutoff valve device **1060** may be a device that suppresses inflow of the refrigerant into the plurality of utilization units **1030** by one first shutoff valve **1052** and/or one second shutoff valve **1054**.
Third Embodiments
(1) Overall Outline

(268) An outline of an air conditioner **2100** according to one or more embodiments will be described with reference to FIGS. **14** and **15**. FIG. **14** is a schematic configuration diagram of the air conditioner **2100**. FIG. **15** is a control block diagram of the air conditioner **2100**. As described later, in one or more embodiments, the air conditioner **2100** includes a plurality of utilization units **2030** each having a second control unit **2038** and a plurality of shutoff valve devices **2060** each having a control unit **2062**. However, in FIG. **15**, only one second control unit **2038** and one control unit **2062** are illustrated in order to avoid complication of the drawing.
(269) Note that the overall outline of the air conditioner **2100** according to one or more embodiments is similar to the overall outline of the air conditioner **1100** according to the above-described embodiments if reference numerals in the **1000** series are replaced with reference numerals in the **2000** series, and thus the description thereof will be omitted here.
(2) Detailed Configuration
(270) The heat source unit **2010**, the utilization unit **2030**, and the shutoff valve device **2060** will be described in detail.
(271) (2-1) Heat Source Unit
(272) The description of the heat source unit **2010** according to one or more embodiments is similar to the description of the heat source unit **1010** according to the above-described embodiments if reference numerals in the **1000** series are replaced with reference numerals in the **2000** series, and thus the description thereof will be omitted here.
(273) (2-2) Utilization Unit
(274) The utilization unit **2030** will be described. In the utilization unit **2030**, heat is exchanged between the refrigerant and the air in the air conditioning target space **2000**R in the utilization heat exchanger **2032** described later, and as a result, cooling or heating of the air conditioning target space **2000**R is performed.
(275) In one or more embodiments, the air conditioner **2100** includes three utilization units **2030**. Structures and capabilities of the three utilization units **2030** may be the same or different from each other. Here, each of the utilization units **2030** will be described as having the same configuration.
(276) The type of the utilization unit **2030** is a floor type as illustrated in FIG. **16**, and is installed on the floor of the air conditioning target space **2000**R. The air conditioner **2100** may include another type of utilization unit **2030** in addition to the floor utilization unit **2030**.
(277) As shown in FIGS. **14** to **16**, the utilization unit **2030** mainly includes a casing **2042**, a utilization heat exchanger **2032**, a second expansion valve **2034**, a second fan **2036**, a refrigerant detector **2040**, and a second control unit **2038**. Note that the configuration of the utilization unit **2030** illustrated here is merely an example. The utilization unit **2030** may not have a part of the illustrated configuration or may have a configuration other than the illustrated configuration as long as the air conditioner **2100** can function.
(278) The utilization unit **2030** includes, as refrigerant pipes, a liquid refrigerant pipe **2037***a* and a gas refrigerant pipe **2037***b* connected to the utilization heat exchanger **2032** (see FIG. **14**). The liquid refrigerant pipe **2037***a* connects the liquid-refrigerant connection pipe **2000**LP and the liquid side of the utilization heat exchanger **2032**. The gas refrigerant pipe **2037***b* connects the gas-refrigerant connection pipe **2000**GP and the gas side of the utilization heat exchanger **2032**. The liquid refrigerant pipe **2037***a* is provided with a second expansion valve **2034**.
(279) (2-2-1) Casing
(280) The casing **2042** accommodates therein various devices of the utilization unit **2030** including the utilization heat exchanger **2032**, the second expansion valve **2034**, and the second fan **2036**. As shown in FIG. **16**, the casing **2042** is disposed in the air conditioning target space **2000**R.
(281) The casing **2042** is provided with a suction port (not shown) for taking in the air in the air conditioning target space **2000**R. The casing **2042** has a blow-out port (not shown) through which air introduced into the casing **2042** through the suction port and having exchanged heat with the

refrigerant in the utilization heat exchanger **2032** is blown out into the air conditioning target space **2000**R.
(282) (2-2-2) Indoor Heat Exchanger
(283) In the utilization heat exchanger **2032**, heat is exchanged between the refrigerant flowing through the utilization heat exchanger **2032** and air. The utilization heat exchanger **2032** functions as an evaporator of the refrigerant during the cooling operation, and functions as a condenser (radiator) of the refrigerant during the heating operation. Although not limited, the utilization heat exchanger **2032** is, for example, a fin-and-tube heat exchanger having a plurality of heat transfer tubes and a plurality of heat transfer fins.
(284) (2-2-3) Second Expansion Valve
(285) The second expansion valve **2034** is a mechanism that decompresses the refrigerant and adjusts the flow rate of the refrigerant. In one or more embodiments, the second expansion valve **2034** is an electronic expansion valve whose opening degree is adjustable. The opening degree of the second expansion valve **2034** is appropriately adjusted according to the operation situation. The second expansion valve **2034** is not limited to the electronic expansion valve, and may be another type of valve such as an automatic temperature expansion valve.
(286) (2-2-4) Second Fan
(287) The second fan **2036** is a blower that generates an air flow that flows into the casing **2042** from a suction port (not shown) of the casing **2042**, passes through the utilization heat exchanger **2032**, and then flows out of the casing **2042** from a blow-out port of the casing **2042**. The second fan **2036** is, for example, an inverter control-type fan. However, the second fan **2036** may be a constant-speed fan.
(288) (2-2-5) Refrigerant Detector
(289) The refrigerant detector **2040** is a sensor that detects the leakage of refrigerant at the utilization unit **2030**. The refrigerant detector **2040** is provided, for example, in the casing **2042** of the utilization unit **2030**. The refrigerant detector **2040** may be disposed outside the casing **2042** of the utilization unit **2030**. A plurality of refrigerant detectors **2040** may be provided.
(290) The refrigerant detector **2040** is, for example, a semiconductor-type sensor. The semiconductor refrigerant detector **2040** includes a semiconductor-type detection element (not shown). The semiconductor detector element has electric conductivity that changes depending on whether it is in a case where no refrigerant gas exists therearound and in a case where refrigerant gas exists therearound. When the refrigerant gas exists around the semiconductor-type detection element, the refrigerant detector **2040** outputs a relatively large current as a detection signal. On the other hand, when the refrigerant gas does not exist around the semiconductor-type detection element, the refrigerant detector **2040** outputs a relatively small current as a detection signal.
(291) The type of the refrigerant detector **2040** is not limited to the semiconductor type, and may be any sensor capable of detecting refrigerant gas. For example, the refrigerant detector **2040** may be an infrared sensor that outputs a detection signal according to a detection result of the refrigerant.
(292) (2-2-6) Second Control Unit
(293) The second control unit **2038** controls operations of various devices of the utilization unit **2030**. The second control unit **2038** includes a microcontroller unit (MCU), various electric circuits, and electronic circuits (not shown). The MCU includes a CPU, a memory, an I/O interface, and the like. The memory of the MCU stores various programs to be executed by the CPU of the MCU. Note that the various functions of the second control unit **2038** does not need to be implemented by software, and may be implemented by hardware or may be implemented by cooperation of hardware and software.
(294) The second control unit **2038** is electrically connected to various devices of the utilization unit **2030** including the second expansion valve **2034** and the second fan **2036** (see FIG. **15**). The second control unit **2038** is electrically connected to the refrigerant detector **2040**. Further, the second control unit **2038** is electrically connected to a sensor (not shown) provided in the

utilization unit **2030**. Although not limited, the sensors (not shown) include a temperature sensor provided in the utilization heat exchanger **2032** and the liquid refrigerant pipe **2037***a*, a temperature sensor that measures the temperature of the air conditioning target space **2000**R, and the like. The utilization unit **2030** may include all or some of these sensors.
(295) The second control unit **2038** is communicably connected to the control unit **2062** that controls the operations of the first shutoff valve **2052** and the second shutoff valve **2054** of the shutoff valve device **2060** by a communication line (see FIG. **15**).
(296) As illustrated in FIG. **15**, the second control unit **2038** is connected to the first control unit **2022** of the heat source unit **2010** via a communication line. The second control unit **2038** is communicably connected to a remote control unit for operating the air conditioner **2100** (not shown) via a communication line. The first control unit **2022** and the second control unit **2038** cooperate to function as a controller **2090** that controls the operation of the air conditioner **2100**.
(297) The function of the controller **2090** will be described. Some or all of various functions of the controller **2090** described below may be executed by a control device provided separately from the first control unit **2022** and the second control unit **2038**.
(298) The controller **2090** controls the operation of the flow direction switching mechanism **2014** such that the heat-source heat exchanger **2016** functions as a condenser of the refrigerant and the utilization heat exchanger **2032** functions as an evaporator of the refrigerant during the cooling operation. During the heating operation, the controller **2090** controls the operation of the flow direction switching mechanism **2014** such that the heat-source heat exchanger **2016** functions as an evaporator of the refrigerant and the utilization heat exchanger **2032** functions as a condenser of the refrigerant. The controller **2090** operates the compressor **2012**, the first fan **2020**, and the second fan **2036** during the cooling operation and the heating operation. During the cooling operation and the heating operation, the controller **2090** adjusts the numbers of rotations of the motors of the compressor **2012**, the first fan **2020**, and the second fan **2036** and the opening degrees of the first expansion valve **2018** and the second expansion valve **2034** based on various instructions (set temperature, set air volume, and the like) input to the remote control unit and measurement values of various temperature sensors and pressure sensors. Various control modes are generally known for the control of the operations of the various devices of the air conditioner **2100** during the cooling operation and the heating operation, and thus, the description thereof will be omitted here.
(299) The control of the air conditioner **2100** by the controller **2090** when the refrigerant is detected by the refrigerant detector **2040** of any of the utilization units **2030** will be described later.
(300) (2-3) Shutoff Valve Device
(301) Each shutoff valve device **2060** is installed corresponding to one of the utilization units **2030**. The shutoff valve device **2060** is a device that closes the shutoff valve **2050** included in the shutoff valve device **2060** to suppress the inflow of the refrigerant into the utilization unit **2030** corresponding to the shutoff valve device **2060**.
(302) The shutoff valve device **2060** mainly includes a shutoff valve **2050**, a main body casing **2064**, an electric component **2062***a*, and an electric component box **2066** that accommodates the electric component **2062***a*. The electric component **2062***a* includes a control unit **2062** that controls operations of the first shutoff valve **2052** and the second shutoff valve **2054**.
(303) (2-3-1) Shutoff Valve
(304) The shutoff valve **2050** includes at least one of a first shutoff valve disposed in the liquid-refrigerant connection pipe **2000**LP and a second shutoff valve disposed in the gas-refrigerant connection pipe **2000**GP. In one or more embodiments, the shutoff valve **2050** of each shutoff valve device **2060** includes both the first shutoff valve **2052** disposed in the liquid-refrigerant connection pipe **2000**LP and the second shutoff valve **2054** disposed in the gas-refrigerant connection pipe **2000**GP.
(305) A liquid-refrigerant connection pipe **2000**LP connecting the heat source unit **2010** and the first shutoff valve **2052** is connected to one end of the first shutoff valve **2052** of the shutoff valve

device **2060**. A liquid-refrigerant connection pipe **2000**LP that connects the liquid refrigerant pipe **2037***a* of the utilization unit **2030** corresponding to the shutoff valve device **2060** and the first shutoff valve **2052** is connected to the other end of the first shutoff valve **2052** of the shutoff valve device **2060**.

(306) A gas-refrigerant connection pipe **2000**GP connecting the heat source unit **2010** and the second shutoff valve **2054** is connected to one end of the second shutoff valve **2054** of the shutoff valve device **2060**. A gas-refrigerant connection pipe **2000**GP that connects the gas refrigerant pipe **2037***b* of the utilization unit **2030** corresponding to the shutoff valve device **2060** and the second shutoff valve **2054** is connected to the other end of the second shutoff valve **2054** of the shutoff valve device **2060**.

(307) The first shutoff valve **2052** and the second shutoff valve **2054** are valves that suppress the leakage of refrigerant into the air conditioning target space **2000**R when the leakage of refrigerant occurs at the utilization unit **2030**. The first shutoff valve **2052** and the second shutoff valve **2054** are, for example, electromagnetic valves capable of switching between a closed state (fully closed) and an open state (fully open). However, the types of the first shutoff valve **2052** and the second shutoff valve **2054** are not limited to the electromagnetic valve, and may be, for example, an electric valve.

(308) The first shutoff valve **2052** and the second shutoff valve **2054** of the shutoff valve device **2060** are normally opened. The normal time here means a time when the second control unit **2038** of the utilization unit **2030** corresponding to the shutoff valve device **2060** has not transmitted a signal instructing the control unit **2062** to close the shutoff valve **2050**.

(309) On the other hand, when the refrigerant detector **2040** of the utilization unit **2030** corresponding to the shutoff valve device **2060** detects the refrigerant, the first shutoff valve **2052** and the second shutoff valve **2054** are closed. Specifically, when the refrigerant detector **2040** of the utilization unit **2030** corresponding to the shutoff valve device **2060** detects the refrigerant and the second control unit **2038** of the corresponding utilization unit **2030** transmits a signal instructing to close the shutoff valve **2050** to the control unit **2062** of the corresponding shutoff valve device **2060**, the control unit **2062** that has received the signal performs control to close the first shutoff valve **2052** and the second shutoff valve **2054**. When the first shutoff valve **2052** and the second shutoff valve **2054** of the shutoff valve device **2060** are closed, the inflow of the refrigerant from the heat source unit **2010**, the pipe connecting the heat source unit **2010** and the first shutoff valve **2052**, and the pipe connecting the heat source unit **2010** and the second shutoff valve **2054** to the utilization unit **2030** corresponding to the shutoff valve device **2060** is suppressed.

(310) As illustrated in FIG. **16**, the shutoff valve **2050** (the first shutoff valve **2052** and the second shutoff valve **2054**) is disposed in the underfloor space **2000**S below the floor **2000**FL of the air conditioning target space **2000**R. The shutoff valve **2050** may be arranged in the underfloor space **2000**S and in the vicinity of the corresponding utilization unit **2030**. The shutoff valve **2050** is disposed, for example, in the underfloor space **2000**S and in the vicinity immediately below the corresponding utilization unit **2030**. However, the arrangement of the shutoff valve **2050** is not limited to the vicinity immediately below the corresponding utilization unit **2030**.

(311) When the lower floor exists below the air conditioning target space **2000**R in the building in which the air conditioner **2100** is installed, the underfloor space **2000**S is a space between the floor **2000**FL of the air conditioning target space **2000**R and the ceiling of the lower floor (one floor below) of the air conditioning target space **2000**R. For example, the underfloor space **2000**S is a space existing between a building material constituting the floor **2000**FL and a concrete framework partitioning a floor where the air conditioning target space **2000**R exists and a floor immediately below that floor. Further, for example, the underfloor space **2000**S may be a space between a concrete framework that partitions a floor where the air conditioning target space **2000**R exists and a floor immediately below that floor, and a ceiling of a floor immediately below the floor where the

air conditioning target space **2000**R exists (an attic space of a floor immediately below). When the air conditioning target space **2000**R is the lowest floor in the building in which the air conditioner **2100** is installed (when there is no lower floor below the air conditioning target space **2000**R), the underfloor space **2000**S is a space between the floor **2000**FL of the air conditioning target space **2000**R and the foundation of the building.
(312) The underfloor space **2000**S is a space partitioned from the air conditioning target space **2000**R by the building material constituting the floor **2000**FL. The fact that the underfloor space **2000**S and the air conditioning target space **2000**R are partitioned does not necessarily mean that both spaces are partitioned in an airtight state, but means that the flow of air between both spaces is at least suppressed by the building material constituting the floor **2000**FL. For example, some air may flow between the underfloor space **2000**S and the air conditioning target space **2000**R via a gap between building materials.
(313) In the air conditioner **2100** of one or more embodiments, since the shutoff valve **2050** is installed in the underfloor space **2000**S, even if the refrigerant leaks around the shutoff valve **2050**, the refrigerant flows not into the air conditioning target space **2000**R but into the underfloor space **2000**S partitioned from the air conditioning target space **2000**R. Since the refrigerant used in the air conditioner **2100** is generally denser than air, the refrigerant is relatively less likely to flow from the underfloor space **2000**S into the air conditioning target space **2000**R above the underfloor space **2000**S. In short, in the air conditioner **2100** according to one or more embodiments, the refrigerant is less likely to flow into the air conditioning target space **2000**R where people are active. Therefore, the air conditioner **2100** has high safety even when, for example, a flammable refrigerant is used.
(314) (2-3-2) Main Body Casing
(315) The main body casing **2064** is a casing that accommodates the shutoff valve **2050**. Specifically, the main body casing **2064** is a casing that accommodates the first shutoff valve **2052** and the second shutoff valve **2054**.
(316) As shown in FIG. **17**A, the main body casing **2064** is installed in the underfloor space **2000**S. In order to dispose the shutoff valve **2050** in the vicinity of the corresponding utilization unit **2030**, the main body casing **2064** may be installed in the vicinity of the corresponding utilization unit **2030**. Although not limited, the main body casing **2064** is installed in the underfloor space **2000**S and in the vicinity immediately below the corresponding utilization unit **2030**.
(317) For example, as illustrated in FIG. **17**A, the main body casing **2064** is formed with an opening **2064***a* through which a liquid-refrigerant connection pipe **2000**LP connected to both ends of the first shutoff valve **2052** and a gas-refrigerant connection pipe **2000**GP connected to both ends of the second shutoff valve **2054** are inserted. In FIG. **17**A, a part of these openings **2064***a* (for example, the opening **2064***a* through which the liquid-refrigerant connection pipe **2000**LP connecting the heat source unit **2010** and the first shutoff valve **2052** and the gas-refrigerant connection pipe **2000**GP connecting the heat source unit **2010** and the second shutoff valve **2054** pass) is illustrated. In the example of FIG. **17**A, a plurality of refrigerant pipes (one liquid-refrigerant connection pipe **2000**LP and one gas-refrigerant connection pipe **2000**GP) are arranged to extend through one opening **2064***a*.
(318) In place of the openings through which the plurality of refrigerant pipes pass, the main body casing **2064** may include, as shown in FIG. **17**B, openings **2064***a* arranged so that one refrigerant pipe (one liquid-refrigerant connection pipe **2000**LP or one gas-refrigerant connection pipe **2000**GP) extends through each of the opening **2064***a*.
(319) The opening **2064***a* may be provided with a heat insulating material **2068** that closes a gap between the opening **2064***a* and the liquid-refrigerant connection pipe **2000**LP, a gap between the opening **2064***a* and the gas-refrigerant connection pipe **2000**GP, and a gap between the liquid-refrigerant connection pipe **2000**LP and the gas-refrigerant connection pipe **2000**GP. By closing the gap between the opening **2064***a* and the connection pipes **2000**LP and **2000**GP and the gap between

the refrigerant pipes with the heat insulating material **2068** in this manner, the leakage of refrigerant into the underfloor space **2000**S is suppressed even if the refrigerant leaks inside the main body casing **2064**, and safety is therefore high.
(320) (2-3-3) Electric Component
(321) The electric component **2062***a* includes various components for operating the first shutoff valve **2052** and the second shutoff valve **2054**. Although not limited, the electric component **2062***a* includes, for example, a switching unit capable of switching the flow of current, such as a printed circuit board, an electromagnetic relay, and a switching element, a terminal block to which power is supplied, and an input unit to which a signal from the second control unit **2038** is input. The electric component **2062***a* is electrically connected to the first shutoff valve **2052** and the second shutoff valve **2054** by an electric wire for supplying a drive voltage. At least a part of the electric component **2062***a* functions as the control unit **2062** that closes the first shutoff valve **2052** and the second shutoff valve **2054** in response to a signal requesting closing of the shutoff valve **2050** from the second control unit **2038** of the utilization unit **2030**.
(322) The control unit **2062** includes, for example, a microcontroller unit (MCU), various electric circuits, and electronic circuits (not shown). The MCU includes a CPU, a memory, an I/O interface, and the like. The memory of the MCU stores various programs to be executed by the CPU of the MCU. Note that the various functions of the control unit **2062** does not need to be implemented by software, and may be implemented by hardware or may be implemented by cooperation of hardware and software.
(323) The electric component **2062***a* is accommodated in the electric component box **2066**. The electric component box **2066** may be disposed outside the main body casing **2064**. The electric component box **2066** is installed, for example, in the underfloor space **2000**S. When the electric component box **2066** and the main body casing **2064** have independent configurations, the electric component box **2066** may not be disposed in the vicinity of the main body casing **2064**. The installation position of the electric component box **2066** may be appropriately determined.
(3) Control of Air Conditioner by Controller at Time of Refrigerant Detection
(324) The control of the air conditioner **2100** by the controller **2090** when the refrigerant is detected by the refrigerant detector **2040** of any of the utilization units **2030** will be described. Here, the case where the refrigerant is detected by the refrigerant detector **2040** means a case where the value of the current output as the detection signal by the refrigerant detector **2040** is larger than a predetermined threshold.
(325) When the refrigerant is detected by the refrigerant detector **2040** of any of the utilization units **2030**, the controller **2090** transmits a signal instructing to close the shutoff valve **2050** of the shutoff valve device **2060** to the control unit **2062** of the shutoff valve device **2060** corresponding to the utilization unit **2030** in which the refrigerant is detected. The signal instructing to close the shutoff valve **2050** may be a contact signal. The control unit **2062** of the shutoff valve device **2060** closes the shutoff valve **2050** (in one or more embodiments, the first shutoff valve **2052** and the second shutoff valve **2054**) based on this signal.
(326) When the refrigerant is detected by the refrigerant detector **2040** of any of the utilization units **2030**, the controller **2090** may notify of the leakage of refrigerant using an alarm (not shown) in addition to transmitting a signal instructing to close the shutoff valve **2050** to the control unit **2062** of the shutoff valve device **2060**.
(327) When the refrigerant is detected by the refrigerant detector **2040** of any of the utilization units **2030**, the controller **2090** may stop the operation of the entire air conditioner **2100** by stopping the operation of the compressor **2012** in addition to transmitting a signal instructing to close the shutoff valve **2050** to the control unit **2062** of the shutoff valve device **2060**.
(328) When the refrigerant is detected by the refrigerant detector **2040** of any of the utilization units **2030**, the controller **2090** may transmit a signal instructing to close the shutoff valve **2050** to the control unit **2062** of the shutoff valve device **2060** corresponding to that utilization unit **2030**

and may further transmit a signal instructing to close the shutoff valve **2050** to the control unit **2062** of another shutoff valve device **2060** (for example, all the shutoff valve devices **2060**).
(4) Features
(329) (4-1)
(330) The air conditioner **2100** of one or more embodiments includes a utilization unit **2030** as an air conditioning indoor unit, a heat source unit **2010** as an air conditioning heat source unit, and a shutoff valve device **2060**. The utilization unit **2030** is installed in the air conditioning target space **2000**R. The utilization unit **2030** is a floor type. The heat source unit **2010** is connected to the utilization unit **2030** via a liquid-refrigerant connection pipe **2000**LP and a gas-refrigerant connection pipe **2000**GP. The shutoff valve device **2060** includes a shutoff valve **2050** disposed in the underfloor space **2000**S below the floor **2000**FL of the air conditioning target space **2000**R. The shutoff valve **2050** includes at least one of a first shutoff valve **2052** disposed in the liquid-refrigerant connection pipe **2000**LP and a second shutoff valve **2054** disposed in the gas-refrigerant connection pipe **2000**GP. In one or more embodiments, the shutoff valve **2050** includes both the first shutoff valve **2052** and the second shutoff valve **2054**.
(331) Conventionally, as disclosed in Patent Literature 2 (JP 2013-19621 A), there is known an air conditioner in which an air conditioning indoor unit is provided with a shutoff valve for preventing the leakage of refrigerant separately from the air conditioning indoor unit. The shutoff valve is disposed in the vicinity of the air conditioning indoor unit to reduce the amount of refrigerant leaking from the air conditioning indoor unit when the leakage of refrigerant is detected.
(332) However, when the shutoff valve is disposed in the air conditioning target space adjacent to the air conditioning indoor unit disposed in the air conditioning target space, if the refrigerant leaks from the shutoff valve, a relatively large amount of refrigerant may leak into the air conditioning target space.
(333) In the air conditioner **2100**, even if the refrigerant leaks around the shutoff valve **2050**, the refrigerant flows not into the air conditioning target space **2000**R but into the underfloor space **2000**S partitioned from the air conditioning target space **2000**R, and thus safety is high.
(334) Since the refrigerant used in the air conditioner **2100** is generally denser than air, the refrigerant flowing into the underfloor space **2000**S is relatively less likely to flow into the air conditioning target space **2000**R.
(335) Further, in the air conditioner **2100**, since the underfloor space **2000**S close to the utilization unit **2030** is used as the installation place of the shutoff valve **2050** for the floor type utilization unit **2030**, the length of the pipe connecting the utilization unit **2030** and the shutoff valve **2050** tends to be relatively short. Therefore, even if the refrigerant leaks from the utilization unit **2030**, the amount of refrigerant leaking from the utilization unit **2030** is easily reduced.
(336) (4-2)
(337) In the air conditioner **2100** of one or more embodiments, the shutoff valve device **2060** includes a main body casing **2064** as an example of a casing that accommodates the first shutoff valve **2052** and the second shutoff valve **2054**.
(338) In the present air conditioner **2100**, the shutoff valve device **2060** is a unit in which the first shutoff valve **2052**, the second shutoff valve **2054**, and the main body casing **2064** accommodating them are unitized. It is therefore easy to incorporate the shutoff valve device **2060** into the air conditioner **2100**.
(339) (4-3)
(340) In the air conditioner **2100** of one or more embodiments, the shutoff valve device **2060** includes an electric component box **2066** that accommodates electric components **2062***a* for operating the shutoff valve **2050**. The electric component box **2066** is disposed outside the main body casing **2064**.
(341) In the present air conditioner **2100**, since the electric component box **2066** is disposed outside the main body casing **2064**, if the refrigerant is flammable, and the refrigerant leaks around

the shutoff valve **2050**, it is possible to suppress contact between the refrigerant and the electric component **2062***a* that can be an ignition source.
(342) (4-4)
(343) In the air conditioner **2100** of one or more embodiments, an opening **2064***a* is formed in the main body casing **2064**. The liquid-refrigerant connection pipe **2000**LP connected to the first shutoff valve **2052** and the gas-refrigerant connection pipe **2000**GP connected to the second shutoff valve **2054** extend through the opening **2064***a* of the main body casing **2064**. The shutoff valve device **2060** includes a heat insulating material **2068** that closes a gap between the opening **2064***a* and the liquid-refrigerant connection pipe **2000**LP and a gap between the opening **2064***a* and the gas-refrigerant connection pipe **2000**GP.
(344) In the present air conditioner **2100**, since the gap between the opening **2064***a* and the refrigerant connection pipes **2000**LP and **2000**GP extending through the opening **2064***a* is closed by the heat insulating material **2068**, the leakage of refrigerant into the underfloor space **2000**S is suppressed even if the refrigerant leaks inside the main body casing **2064**, and safety is high.
(5) Modifications
(345) The above described embodiments may be appropriately modified as described in the following modifications. Some or all of the modifications may be combined with the above-described embodiments or another modification as long as no contradiction occurs.
(346) (5-1) Modification 3A
(347) In the above-described embodiments, the control unit **2062** of the shutoff valve device **2060** controls the operation of the shutoff valve **2050**, but the present disclosure is not limited to such a configuration. For example, the shutoff valve device **2060** may not include the control unit **2062**, and the controller **2090** of the air conditioner **2100**, more specifically, the second control unit **2038** of the utilization unit **2030**, for example, may control the operation of the shutoff valve **2050**.
(348) (5-2) Modification 3B
(349) In the above-described embodiments, the shutoff valve device **2060** includes, as the shutoff valve **2050**, the first shutoff valve **2052** and the second shutoff valve **2054** dedicated for preventing refrigerant leakage. However, in the shutoff valve device **2060**, a valve used for a purpose other than the purpose of preventing refrigerant leakage may be used as the shutoff valve **2050**.
(350) For example, the shutoff valve device **2060***a* of the air conditioner **2100** illustrated in FIG. **18** does not include the first shutoff valve **2052**. The utilization unit **2030***a* of the air conditioner **2100** illustrated in FIG. **18** does not include the second expansion valve **2034**, and instead, the shutoff valve device **2060***a* includes the second expansion valve **2034** as the shutoff valve **2050**. In short, the shutoff valve device **2060***a* includes the second expansion valve **2034** and the second shutoff valve **2054** as the shutoff valve **2050**.
(351) In the air conditioner **2100** illustrated in FIG. **18**, the controller **2090** of the air conditioner **2100** also functions as a control unit of the shutoff valve device **2060***a*. However, the present disclosure is not limited to such a configuration, and the shutoff valve device **2060***a* may include a control unit that controls operations of the second expansion valve **2034** and the second shutoff valve **2054**. During the cooling operation and the heating operation, the controller **2090** adjusts the opening degree of the second expansion valve **2034** based on various instructions (set temperature, set air volume, and the like) input to the remote control unit and measurement values of various temperature sensors and pressure sensors. When the refrigerant is detected by the refrigerant detector **2040** of any of the utilization units **2030**, the controller **2090** closes the second expansion valve **2034** and the second shutoff valve **2054** of the shutoff valve device **2060** corresponding to the utilization unit **2030** in which the refrigerant is detected.
(352) (5-3) Modification 3C
(353) In the above-described embodiments, the shutoff valve device **2060** includes, as the shutoff valve **2050**, both the first shutoff valve **2052** disposed in the liquid-refrigerant connection pipe **2000**LP and the second shutoff valve **2054** disposed in the gas-refrigerant connection pipe **2000**GP.

However, as illustrated in FIG. **19**, the shutoff valve device **2060***b* of the air conditioner **2100** may include only the second shutoff valve **2054** as the shutoff valve **2050**.
(354) Here, when the refrigerant is detected by the refrigerant detector **2040** of any of the utilization units **2030**, the controller **2090** transmits an instruction signal to the control unit **2062** of the shutoff valve device **2060***b* corresponding to the utilization unit **2030** in which the refrigerant is detected so as to close the second shutoff valve **2054**. Further, when the refrigerant detector **2040** of any of the utilization units **2030** detects the refrigerant, the controller **2090** may close the second expansion valve **2034** of the utilization unit **2030** in which the refrigerant is detected.
(355) (5-4) Modification 3D
(356) In the above-described embodiments, the shutoff valve device **2060** includes the main body casing **2064** that accommodates the shutoff valve **2050** therein, but the present disclosure is not limited thereto. The shutoff valve device **2060** may not include the main body casing **2064**, and the shutoff valve **2050** may be disposed in the underfloor space **2000**S as it is.
(357) In addition, the electric component **2062***a* may also be disposed in the underfloor space **2000**S as it is instead of in the electric component box **2066**.
(358) (5-5) Modification 3E
(359) In the above-described embodiments, one shutoff valve device **2060** is provided for each utilization unit **2030**, but the present disclosure is not limited thereto. For example, the shutoff valve device **2060** may be a device in which the shutoff valves **2050** for the plurality of utilization units **2030** are accommodated in one main body casing **2064**.
(360) (5-6) Modification 3F
(361) In the above-described embodiments, one first shutoff valve **2052** and one second shutoff valve **2054** are provided for each utilization unit **2030**, but the present disclosure is not limited thereto.
(362) For example, in the air conditioner, one first shutoff valve and one second shutoff valve may be provided in each of the liquid refrigerant pipe and the gas refrigerant pipe before being branched to supply the refrigerant to the plurality of utilization units **2030** (referred to as a utilization unit group). When the refrigerant is detected by one refrigerant detector **2040** of the utilization units belonging to the utilization unit group, the inflow of the refrigerant into the plurality of utilization units **2030** belonging to the utilization unit group may be suppressed by closing the first shutoff valve and the second shutoff valve. In other words, the shutoff valve device **2060** may be a device that suppresses inflow of the refrigerant into the plurality of utilization units **2030** by one first shutoff valve **2052** and/or one second shutoff valve **2054**.
Supplementary Note
(363) Although the disclosure has been described with respect to only a limited number of embodiments, those skilled in the art, having benefit of this disclosure, will appreciate that various other embodiments may be devised without departing from the scope of the present disclosure. Accordingly, the scope of the disclosure should be limited only by the attached claims.
(364) The present disclosure is widely applicable and useful to an air conditioner including a shutoff valve for preventing refrigerant leakage and an air conditioning indoor unit used in the air conditioner.
REFERENCE SIGNS LIST
(365) **30**, **30***a*, **130**: air conditioning indoor unit **32**, **132**: utilization heat exchanger (heat exchanger) **34**: second expansion valve (first shutoff valve) **37***a*: liquid refrigerant pipe **37***b*: gas refrigerant pipe **40**, **140**: casing **46***a*: suction port (opening) **46***b*: blow-out port (opening) **52**: first shutoff valve **54**: second shutoff valve **60**, **160**: partition wall **100**, **100***a*: air conditioner **144***a*: suction opening **144***b*: blow-out opening **481**: suction opening (opening) **482**: blow-out opening (opening) R: air conditioning target space S**1**: first space S**2**: second space **1010**: heat source unit (air conditioning heat source unit) **1030**, **1030***a*: utilization unit (air conditioning indoor unit) **1034**: second expansion valve (first shutoff valve) **1050**: shutoff valve **1052**: first shutoff valve **1054**:

second shutoff valve **1060**, **1060***a*, **1060***b*: shutoff valve device **1062***a*: electric component **1064**: main body casing (casing) **1064***a*: opening **1066**: electric component box **1068**: heat insulating material **1100**: air conditioner **1000**CL: ceiling **1000**GP: gas-refrigerant connection pipe **1000**LP: liquid-refrigerant connection pipe **1000**R: air conditioning target space **1000**S: attic space **2010**: heat source unit (air conditioning heat source unit) **2030**, **2030***a*: utilization unit (air conditioning indoor unit) **2034**: second expansion valve (first shutoff valve) **2050**: shutoff valve **2052**: first shutoff valve **2054**: second shutoff valve **2060**, **2060***a*, **2060***b*: shutoff valve device **2062***a*: electric component **2064**: main body casing (casing) **2064***a*: opening **2066**: electric component box **2068**: heat insulating material **2100**: air conditioner **2000**FL: floor **2000**GP: gas-refrigerant connection pipe **2000**LP: liquid-refrigerant connection pipe **2000**R: air conditioning target space **2000**S: underfloor space
PATENT LITERATURE
(366) Patent Literature 1: JP 2018/011994 W Patent Literature 2: JP 2013-19621 A

Claims

1. An air conditioning indoor unit that blows air, which has exchanged heat with a refrigerant flowing through a heat exchanger, into an air conditioning target space, the air conditioning indoor unit comprising: a liquid refrigerant pipe and a gas refrigerant pipe that are connected to the heat exchanger; a casing that accommodates the heat exchanger and that has an opening communicating with the air conditioning target space; a first shutoff valve in the liquid refrigerant pipe; a second shutoff valve in the gas refrigerant pipe; and a partition wall that extends from a bottom plate of the casing to a top panel of the casing and divides a space inside the casing into a first space and a second space, wherein the first shutoff valve and the second shutoff valve are in the first space, and the second space communicates with the air conditioning target space via the opening.
2. The air conditioning indoor unit according to claim 1, wherein the air conditioning indoor unit is a ceiling-mounted type.
3. The air conditioning indoor unit according to claim 2, wherein the air conditioning indoor unit is a ceiling-embedded type.
4. The air conditioning indoor unit according to claim 2, wherein the first space communicates with an attic space.
5. An air conditioner comprising the air conditioning indoor unit according to claim 1.
6. An air conditioner comprising: an air conditioning indoor unit in an air conditioning target space; an air conditioning heat source unit connected to the air conditioning indoor unit via a liquid refrigerant pipe and a gas refrigerant pipe; and a shutoff valve device comprising a shutoff valve group disposed in an attic space above a ceiling of the air conditioning target space, wherein the shutoff valve group comprises: a first shutoff valve in the liquid refrigerant pipe; and a second shutoff valve in the gas refrigerant pipe, the shutoff valve device further comprises a casing that accommodates the shutoff valve group, the casing has an opening through which the liquid refrigerant pipe and the gas refrigerant pipe extend, and the shutoff valve device further comprises a heat insulating material that closes only a gap between the opening and the liquid refrigerant pipe and a gap between the opening and the gas refrigerant pipe.
7. The air conditioner according to claim 6, wherein the air conditioning indoor unit is a wall-mounted type.
8. The air conditioner according to claim 6, wherein the air conditioning indoor unit is a floor type.
9. The air conditioner according to claim 6, wherein the air conditioning indoor unit is a ceiling suspended type.
10. The air conditioner according to claim 6, wherein the shutoff valve device further comprises an electric component box that accommodates electric components that operates the shutoff valve group, and the electric component box is disposed outside the casing.

11. An air conditioner comprising: a floor-type air conditioning indoor unit in an air conditioning target space; an air conditioning heat source unit connected to the floor-type air conditioning indoor unit via a liquid refrigerant pipe and a gas refrigerant pipe; and a shutoff valve device comprising a shutoff valve group disposed in an underfloor space that is between a floor of the air conditioning target space and a ceiling of a floor immediately below the floor of the air conditioning target space, wherein the shutoff valve group comprises one or both of: a first shutoff valve in the liquid refrigerant pipe; and a second shutoff valve in the gas refrigerant pipe.